



THE UNIVERSITY OF  
MELBOURNE

# Probability Reliability and Quality

## Design of Experiments

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Kevin Otto

# Schedule and Learning Outcomes

| Week | Recorded Lecture Topic                         |
|------|------------------------------------------------|
| 1    | Introduction.                                  |
| 2    | Inspection. Yield.                             |
| 3    | ANOVA. Measurement Systems Analysis.           |
| 4    | Statistics. Confidence Intervals.              |
| 5    | Hypothesis Testing. Sample Sizes.              |
| 6    | Quality Management Processes.                  |
| 7    | Regression.                                    |
| 8    | <b>Design of Experiments.</b>                  |
| 9    | System Reliability. Availability.              |
|      | <i>BREAK – Non Teaching Week</i>               |
| 10   | Parametric Distribution Analysis               |
| 11   | Statistical Process Control. Supplier Quality. |
| 12   | Review                                         |

After completion of the course the student should be able to:

- Explain different statistical methods
- Parametric and non-parametric hypothesis tests
- **Design and plan for collecting experimental data**
- Concepts of reliability engineering and the relationship between TTF, R, H
- How to improve its processes, products and services
- Apply TQM, 6S, and benchmarking



# Overview

Experimentation

Effects and Interactions

Full Factorial Experiment

Fractional Factorial Designs

Center Points

Central Composite Designs

Computer Experiments and Latin Hypercubes



# Process Experimentation

You completed a process map, and FMEA and have a list of variables.

Now you want to experiment with them.

Here we will discuss active experimentation.

Making explicit direct changes to the process to measure what happens.

# Increasing Performance

Decide how to experiment with it

What will you change?

- 1 or 2 factors is easily manageable
- 3 or more factors gets complex



Do not jump to conclusions on things to change serially.

- Make a list of possible causes / factors
- Ask others “why is it performing like this?”



# Increasing Performance

Define 2 or 3 sizes on the 1 or 2 factors

Gear: #1, #2

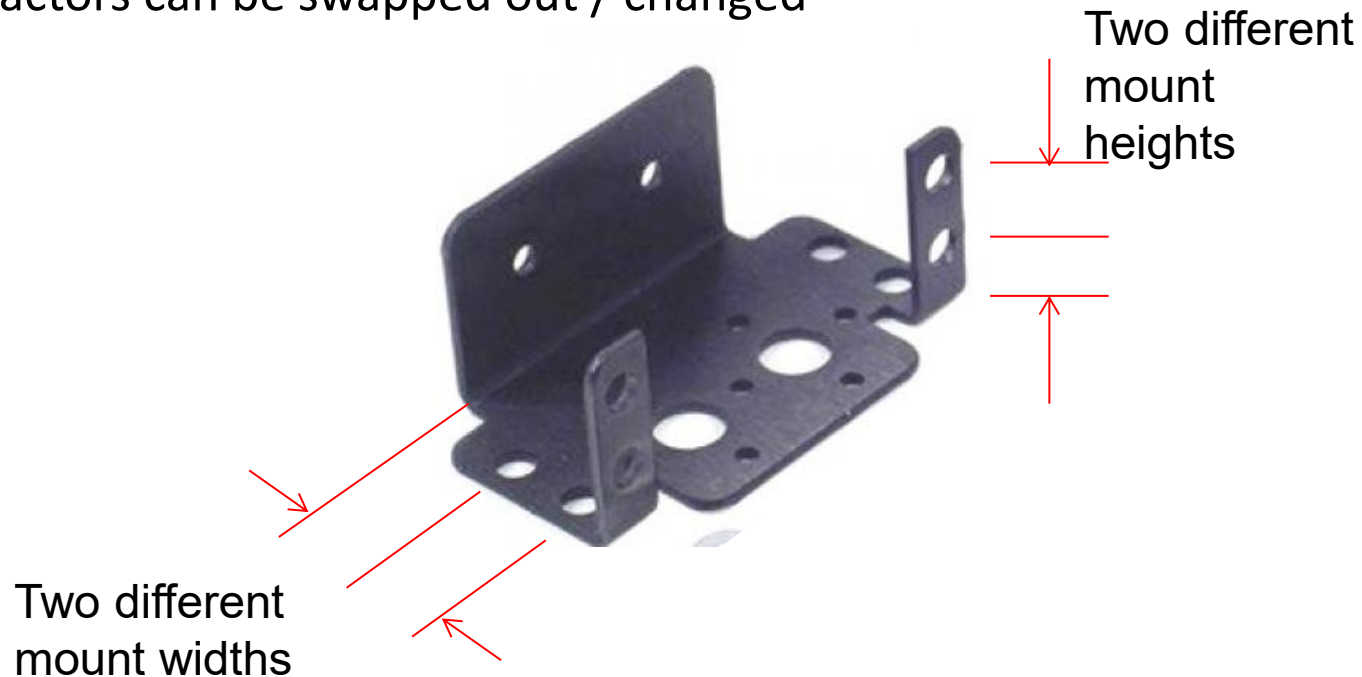
Dimension: 15mm, 17mm, 20mm



# Increasing Performance

Build adjustability into your experimental prototypes

- Where these factors can be swapped out / changed



# Increasing Performance

Define an experimental matrix

- An input  $X$  that you change
- An output  $Y$  that you measure

| $X_1$ | $Y$  |
|-------|------|
| -1    | 2.0  |
| 0     | 6.0  |
| +1    | 12.0 |





# Two Factors

What if changing X didn't do anything?

| $X_1$ | Y   |
|-------|-----|
| -1    | 5.0 |
| 0     | 5.1 |
| +1    | 5.0 |



# Two Factors

What if changing X didn't do anything?

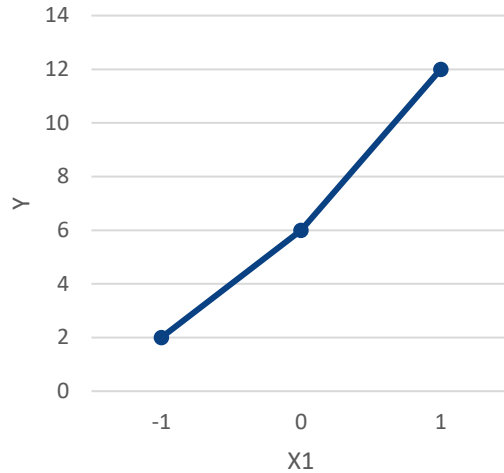
Try changing something else...

| $X_1$ | Y   |
|-------|-----|
| -1    | 5.0 |
| 0     | 5.1 |
| +1    | 5.0 |

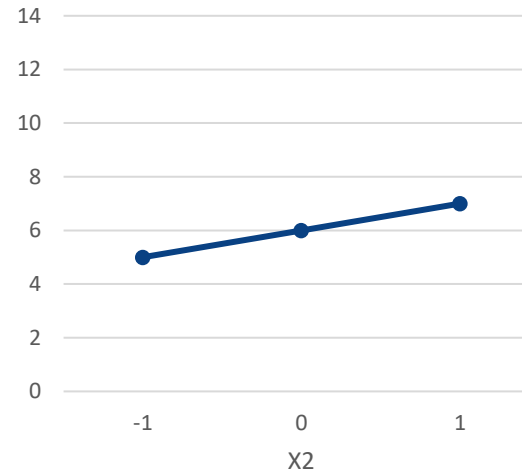
| $X_2$ | Y   |
|-------|-----|
| -1    | 5.0 |
| 0     | 6.0 |
| +1    | 7.0 |

# Two Factors

Graph them. Which factor is more significant?



| $X_1$ | $Y$ |
|-------|-----|
| -1    | 2   |
| 0     | 6   |
| +1    | 12  |



| $X_2$ | $Y$ |
|-------|-----|
| -1    | 5   |
| 0     | 6   |
| +1    | 7   |



# Experimentation

## Summary

- When optimizing performance experimentally
- Clarify a measurable response
- Clarify possible factors to change
  - Look at root causes. Ask why. Ask again.
- Pick one or two factors to change. Do the experiments.
- Graph the results



# Experimentation

## Summary

- When optimizing performance experimentally
- Clarify a measurable response
- Clarify possible factors to change
  - Look at root causes. Ask why. Ask again.
- Pick one or two factors to change. Do the experiments.
- Graph the results

Sometimes you want to use several factors

- Then design of experiments is available to consider



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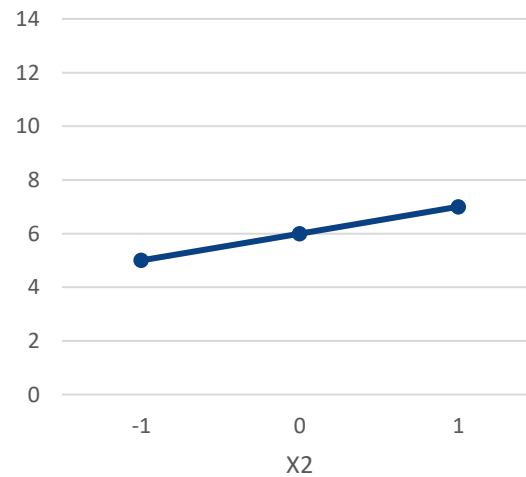
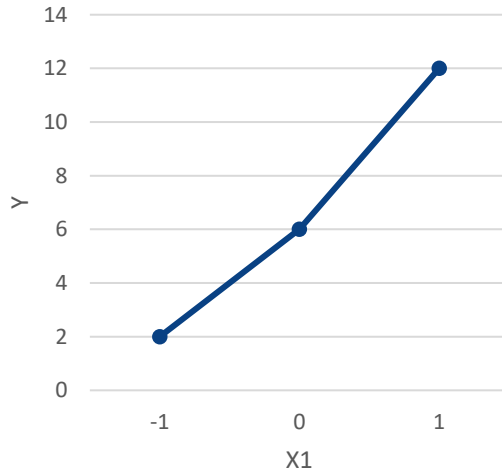
Computer Experiments and Latin Hypercubes

# Two Factors

What if changing X didn't do anything?

Try changing something else...

Most engineers try one factor at a time. OFAT testing.



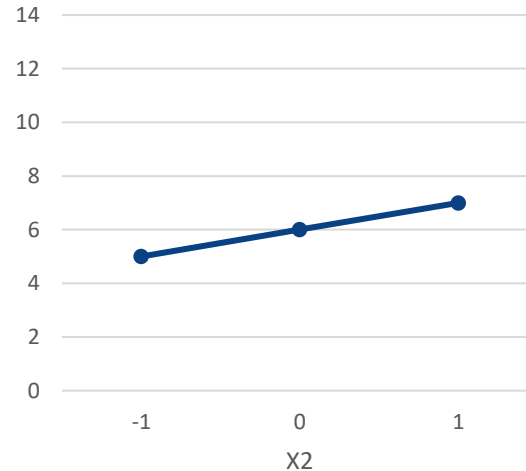
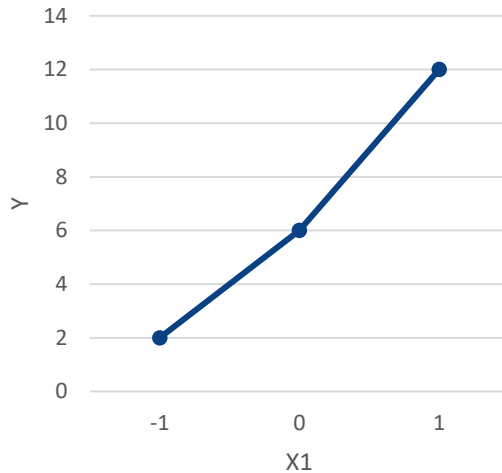
# Experimentation

What if changing X didn't do anything?

Try changing something else...

Sometimes you need to use several factors

- Then design of experiments is available to consider





# Interactions

This example demonstrates a weakness of OFAT testing

- Assume two factors A and B each at two levels, -1 and +1 and a larger-the-better response

|          |    |             |            |
|----------|----|-------------|------------|
| Factor A | +1 | Run 2       | Run 3      |
|          | -1 | Run 1<br>80 | Not Tested |
|          |    | -1          | +1         |
|          |    | Factor B    |            |

# Interactions

This example demonstrates a weakness of OFAT testing

- Assume two factors A and B each at two levels, -1 and +1 and a larger-the-better response

|          |    |             |            |
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| Factor A | +1 | Run 2<br>86 | Run 3      |
|          | -1 | Run 1<br>80 | Not Tested |
|          |    | -1          | +1         |
|          |    | Factor B    |            |

Assume we first vary factor A

- Runs 1 and 2 indicate that  $A^+B^-$  is better than  $A^-B^-$ 
  - So A is held high

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# Interactions

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- Assume two factors A and B each at two levels, -1 and +1 and a larger-the-better response

|          |    |                    |                    |
|----------|----|--------------------|--------------------|
| Factor A | +1 | Run 2<br><b>86</b> | Run 3<br><b>82</b> |
|          | -1 | Run 1<br><b>80</b> | Not Tested         |
|          |    | -1                 | +1                 |
|          |    | Factor B           |                    |

Assume we first vary factor A

- Runs 1 and 2 indicate that  $A^+B^-$  is better than  $A^-B^-$ 
  - So A is held high
  - Next B is changed
- Run 3 indicates that  $A^+B^-$  is better than  $A^+B^+$

# Interactions

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|          |    |                    |                    |
|----------|----|--------------------|--------------------|
| Factor A | +1 | Run 2<br><b>86</b> | Run 3<br><b>82</b> |
|          | -1 | Run 1<br><b>80</b> | Not Tested         |
|          |    | -1                 | +1                 |
|          |    | Factor B           |                    |

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We believe we have found the best point ( $A^+B^-$ ) and stop testing

# Interactions

This example demonstrates a weakness of OFAT testing

- Assume two factors A and B each at two levels, -1 and +1 and a larger-the-better response

| Factor A | +1 | Run 2<br><b>86</b> | Run 3<br><b>82</b>      |
|----------|----|--------------------|-------------------------|
|          | -1 | Run 1<br><b>80</b> | Not Tested<br><b>92</b> |
|          |    | -1                 | +1                      |
|          |    | Factor B           |                         |

Assume we first vary factor A

- Runs 1 and 2 indicate that  $A^+B^-$  is better than  $A^-B^-$ 
  - So A is held high
  - Next B is changed
- Run 3 indicates that  $A^+B^-$  is better than  $A^+B^+$

We believe we have found the best point ( $A^+B^-$ ) and stop testing

In fact,  $A^-B^+$  is best

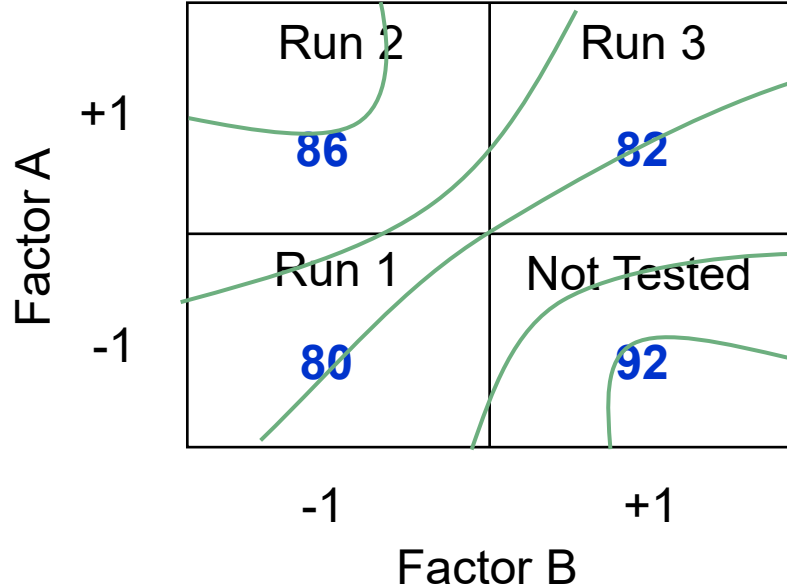
- This is due to the interaction between A and B
- The actual model is

$$Y = 85 - A + 2B - 4AB$$

# Interactions

This example demonstrates a weakness of OFAT testing

- Assume two factors A and B each at two levels, -1 and +1 and a larger-the-better response



Assume we first vary factor A

- Runs 1 and 2 indicate that  $A^+B^-$  is better than  $A^-B^-$ 
  - So A is held high
  - Next B is changed
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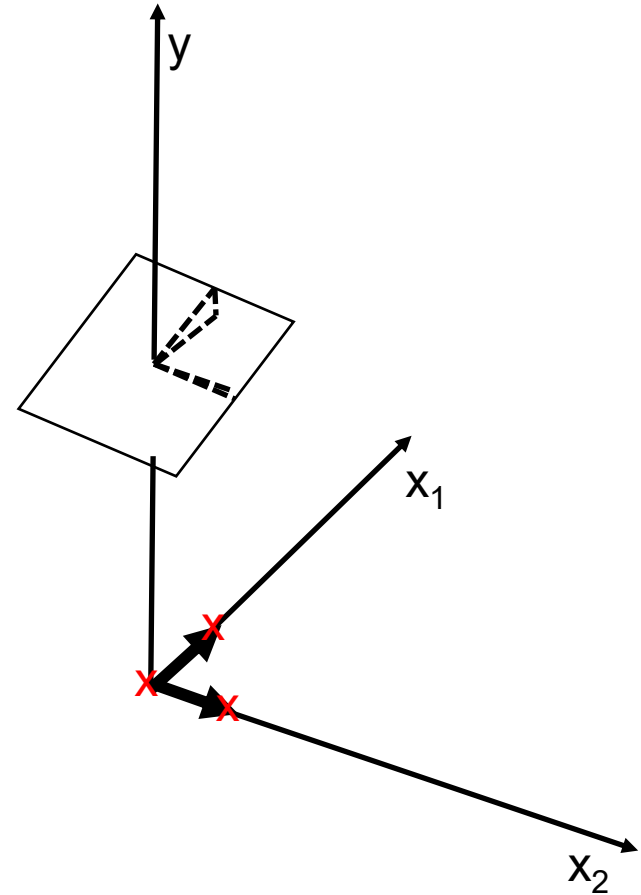
- This is due to the interaction between A and B
- The actual model is

$$Y = 85 - A + 2B - 4AB$$

# OFAT Exploration

OFAT approach is to vary each variable individually

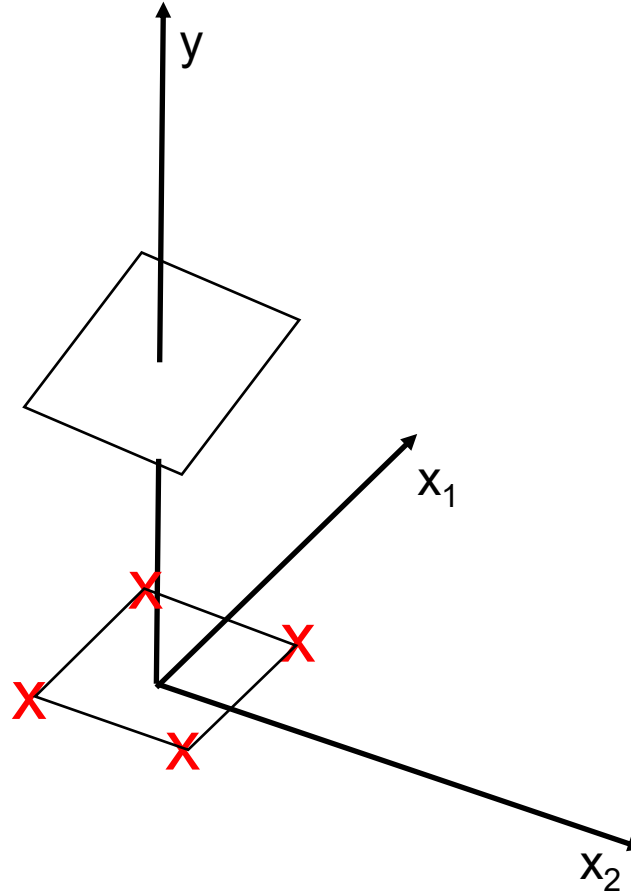
- Vary each from a nominal configuration.





# Corner Points

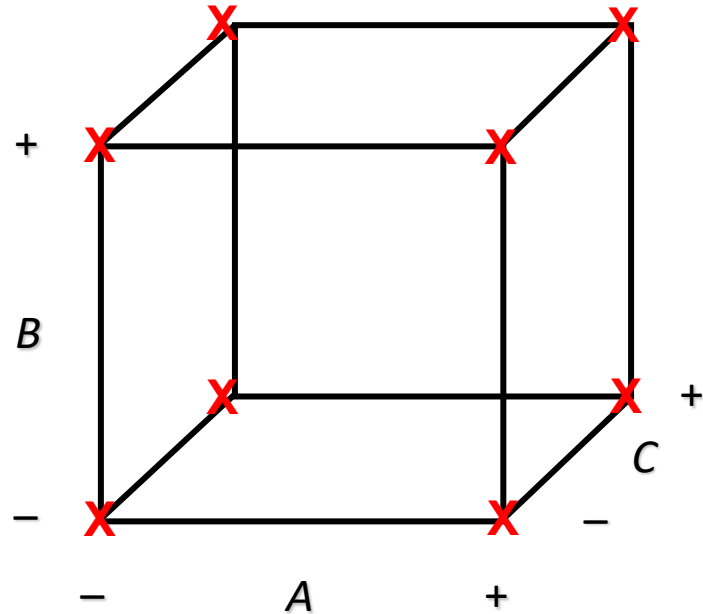
To analyze for interactions, we need to explore the corner locations of the search space.



# Full Factorial

One simple approach is to consider all permutations

This is the *full factorial* experimentation.





# Full Factorial Summary

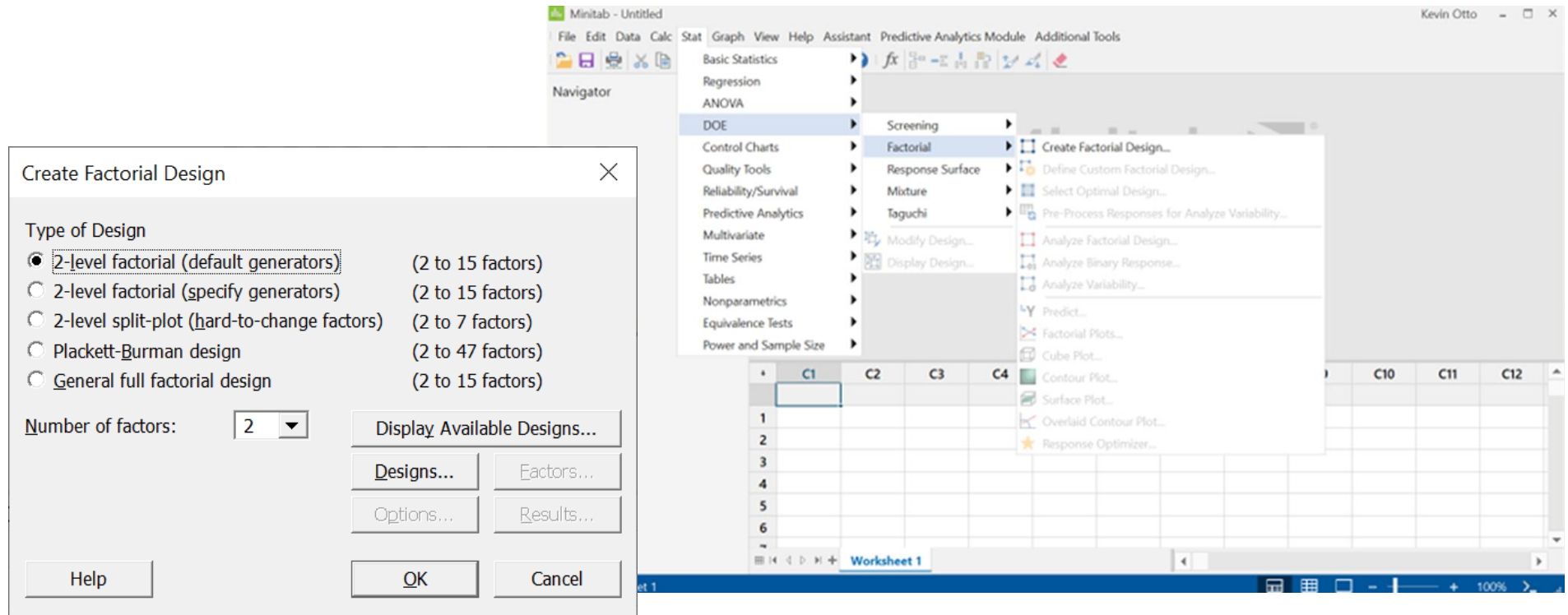
A ***full factorial analysis*** considers all combinations of levels of variables

To do this

- List the variables
- Define levels to consider on each variable
- Enumerate all permutations into a list
- Evaluate the response at all permutations
- Analyze each factor for their effect

There is a menu selection to create various types of designed experiments

Statistics > DOE > Create Factorial Design



The screenshot displays the Minitab software interface. In the foreground, the 'Create Factorial Design' dialog box is open. It features a 'Type of Design' section with five radio button options: '2-level factorial (default generators)' (selected), '2-level factorial (specify generators)', '2-level split-plot (hard-to-change factors)', 'Plackett-Burman design', and 'General full factorial design'. Each option is followed by its factor range in parentheses. Below this, the 'Number of factors' is set to 2. There are buttons for 'Display Available Designs...', 'Designs...', 'Factors...', 'Options...', 'Results...', 'Help', 'OK', and 'Cancel'.

In the background, the Minitab menu bar is visible. The 'Stat' menu is open, and the 'DOE' (Design of Experiments) option is selected. The 'DOE' submenu is also open, showing options like 'Screening', 'Factorial', 'Response Surface', 'Mixture', and 'Taguchi'. The 'Factorial' option is further expanded, showing a list of actions including 'Create Factorial Design...', 'Define Custom Factorial Design...', 'Select Optimal Design...', 'Pre-Process Responses for Analyze Variability...', 'Analyze Factorial Design...', 'Analyze Binary Response...', 'Analyze Variability...', 'Predict...', 'Factorial Plots...', 'Cube Plot...', 'Contour Plot...', 'Surface Plot...', 'Overlaid Contour Plot...', and 'Response Optimizer...'.

# Python or Matlab

Full factorial arrays are easily created in python pyDOE or matlab (identical syntax)

$X = \text{fullfact}([l_1, l_2, \dots, l_n])$

- where  $l_i$  are the number of levels for variable  $i$ .
- The results are in 'coded units' 1,2,...

```
>> X = fullfact([3,4])
```

```
X =  
  1    1  
  2    1  
  3    1  
  1    2  
  2    2  
  3    2  
  1    3  
  2    3  
  3    3  
  1    4  
  2    4  
  3    4
```

# Python or Matlab

Full factorial arrays are easily created in python pyDOE or matlab (identical syntax)

```
X = fullfact([l1,l2,...,ln])
```

- where  $l_i$  are the number of levels for variable  $i$ .
- The results are in 'coded units' 1,2,...
- Create a second 'uncoded units' matrix replacing these with the levels of each variable

```
>> X = fullfact([3,4])
```

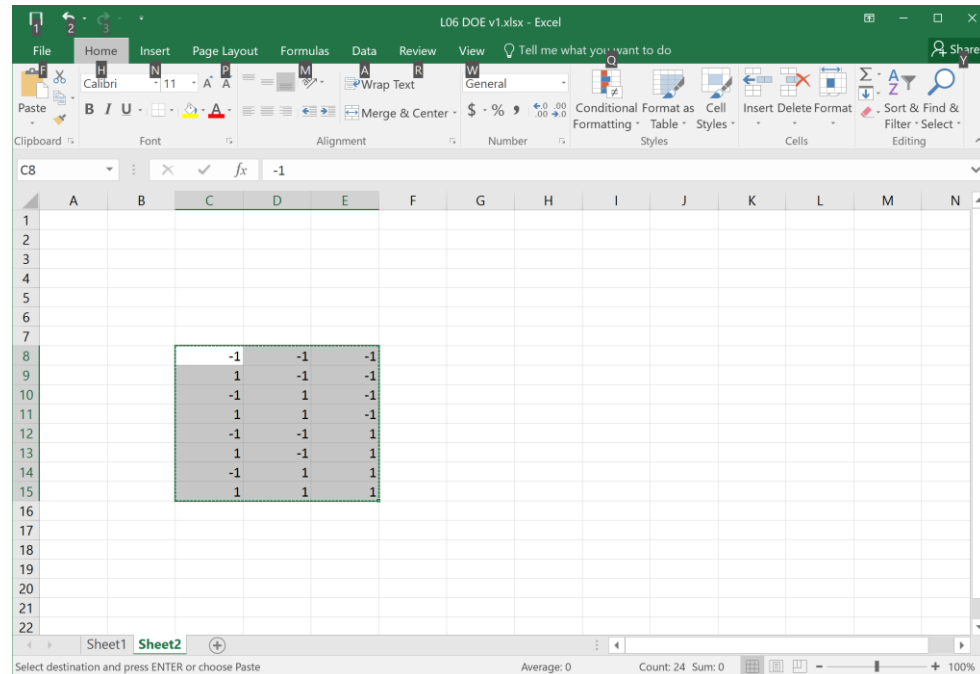
```
X =  
 1  1  
 2  1  
 3  1  
 1  2  
 2  2  
 3  2  
 1  3  
 2  3  
 3  3  
 1  4  
 2  4  
 3  4
```

```
XX =  
 100  0.1  
 150  0.1  
 200  0.1  
 100  0.5  
 150  0.5  
 200  0.5  
 100  0.7  
 150  0.7  
 200  0.7  
 100  0.9  
 150  0.9  
 200  0.9
```

# Excel

In Excel, the arrays must be created manually

- Create, copy and paste blocks
- Unsuitable to scripting and automation





# Design of Experiments

A **factor** (or input, variable) is one of the variables whose influence on a **response** (output) is under investigation

A **level** is one of the values of a factor

- Each chosen value becomes a level
- If the experiment is to be conducted at two different temperatures, then the factor of temperature has two “levels”.

A **treatment combination** is a combination of levels of the factors.

- Example: Pressure = +1, Temperature = -1





# Design of Experiments

A **run** (or trial) is an individual measurement of the response.

- A row in the experimental matrix
- With repeats and replicates, there can be several runs for any treatment combination

A **block** is a set of runs done back-back.

- Often different blocks can be due to experiments done on different days
- Often different blocks can be due to experiments done with different batches of material

The systematic, formal plan for conducting the experiment is called the ***design matrix***.

- It includes the choices of the responses, factors, levels, treatments, blocks



# Design of Experiments

A ***repeat*** is a repeated measurement of a treatment combination

- Take a measurement. Immediately take another. etc.
- Differences in measurements are the *repeatability* variations.
- Provides understanding of experimental error variation.

A ***replicate*** is a reproduced measurement of a treatment combination

- Setup. Measure. Take down. Set up. Measure. Take down. etc.
- Done with different operators or equipment each measurement.
- Differences in measurements are the *reproducibility* variation.
- Provides understanding of experimental setup error.

# Full Factorial

Consider an experiment to determine the effect of Pressure and Temperature on a product's energy efficiency.

- The response?
- Number of variables?
- Number of levels?





## 2<sup>2</sup> Factorial Design

The energy efficiency data collected is the following:

|        | Pressure 1 | Pressure 2 |
|--------|------------|------------|
| Temp 2 | 74         | 82         |
| Temp 1 | 89         | 95         |

## 2<sup>2</sup> Factorial Design

The “Design Matrix” using normalized levels of the factors:

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |

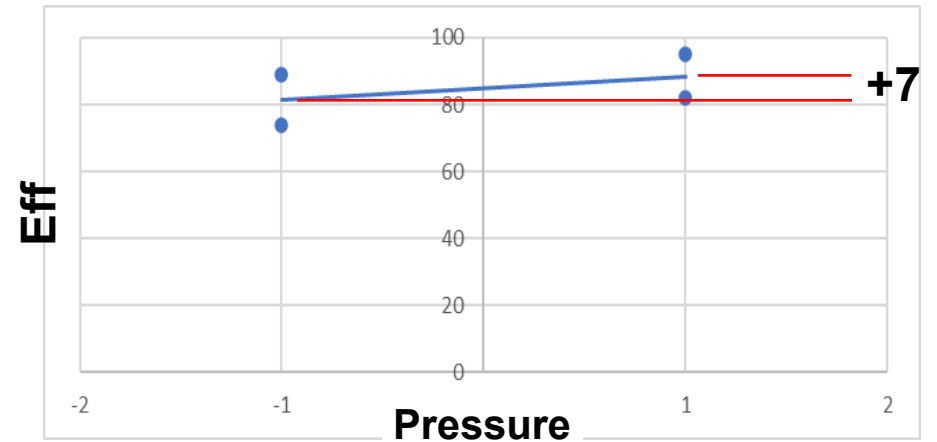
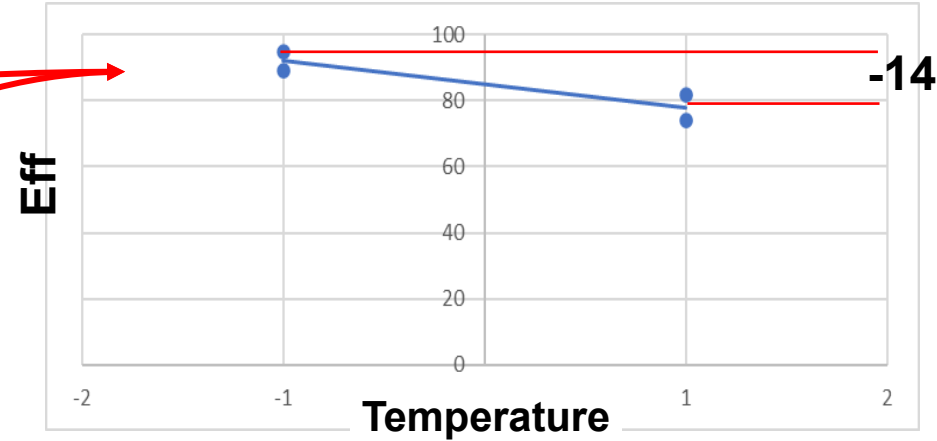
|        | Pressure 1 | Pressure 2 |
|--------|------------|------------|
| Temp 2 | 74         | 82         |
| Temp 1 | 89         | 95         |

What are the “effects” of Temp & Pressure on Efficiency?

# 2<sup>2</sup> Factorial Design

Suppose we just graph Y vs each X

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |



# Main Effects

The change in Efficiency across Temperatures (the Main Effect for Temp) is defined as:

$$ME_{\text{temp}} = \frac{74 + 82}{2} - \frac{89 + 95}{2} = -14$$

As Temperature increases from level -1 to level +1, Efficiency decreases by 14 points.

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |

# Main Effects

The **Main Effect** of a factor is defined as the average change in the output variable produced by a change in the levels of a factor.

The change in Efficiency across Pressures (the Main Effect for Pressure) is defined as:

$$ME_{\text{press}} = \frac{95 + 82}{2} - \frac{89 + 74}{2} = +7$$

As Pressure increases from level -1 to level +1, Efficiency increases by 7 points

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |





# Effects Table

To determine the EFFECT of each factor

- Add up the response values when the factor level is -1
- Add up the response values when the factor level is +1
- Subtract the -1 sum from the +1 sum
- Divide by 2 to determine the EFFECT

|            | Temp | Pressure |  | Efficiency |
|------------|------|----------|--|------------|
|            | -1   | -1       |  | 89         |
|            | 1    | -1       |  | 95         |
|            | -1   | 1        |  | 74         |
|            | 1    | 1        |  | 82         |
| Total –    | 184  | 163      |  |            |
| Total +    | 156  | 177      |  |            |
| Difference | -28  | 14       |  |            |
| Effect     | -14  | 7        |  |            |

# Interactions

Sometimes we find that when fixing one factor, the effect between the levels of another factor is not the same as it's overall main effect.

In this case we  
have an  
***interaction***  
between factors.

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |

$$\left. \begin{array}{c} \text{ } \end{array} \right\} \Delta E_{T-1}(\text{Press}) = 6$$

$$\left. \begin{array}{c} \text{ } \end{array} \right\} \Delta E_{T+1}(\text{Press}) = 8$$

# Interactions

The Temp x Press interaction is the average difference between:

- The effect of Pressure on Efficiency at the low level of Temperature

$$95 - 89 = 6$$

– Note: When the Pressure goes up the Efficiency goes up 6 points

- The effect of Pressure on Efficiency at low level of Temperature

$$82 - 74 = 8$$

– Note: When the Pressure goes up the Efficiency goes up 8 points

$$IE_{\text{Press} \times \text{Temp}} = \frac{(95 - 89) + (82 - 74)}{2} = 1$$

# Effects Table

To determine the EFFECT of each factor

- Add up the response values when the factor level is -1
- Add up the response values when the factor level is +1
- Subtract the -1 sum from the +1 sum, divide by 2 to determine the EFFECT

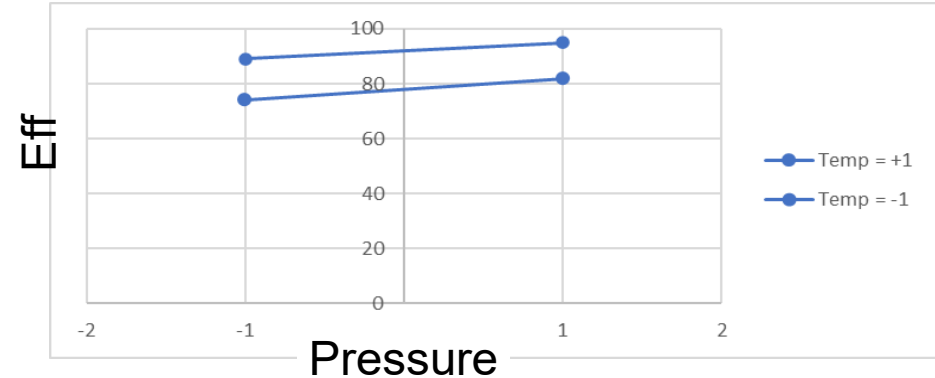
|            | Temp | Pressure | T*P | Efficiency |
|------------|------|----------|-----|------------|
|            | -1   | -1       | 1   | 89         |
|            | 1    | -1       | -1  | 95         |
|            | -1   | 1        | -1  | 74         |
|            | 1    | 1        | 1   | 82         |
| Total –    | 184  | 163      | 169 |            |
| Total +    | 156  | 177      | 171 |            |
| Difference | -28  | 14       | 2   |            |
| Effect     | -14  | 7        | 1   |            |

# Interaction Effect Graph

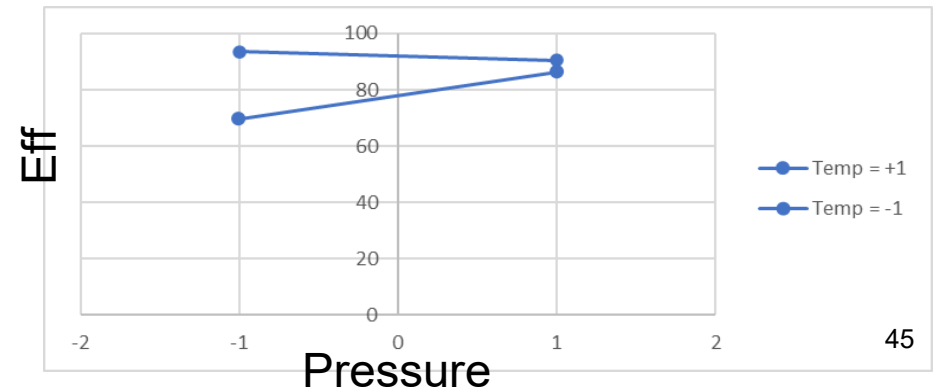
We can graph the interaction

| Temperature | Pressure | Efficiency |
|-------------|----------|------------|
| -1          | -1       | 89         |
| -1          | 1        | 95         |
| 1           | -1       | 74         |
| 1           | 1        | 82         |

Parallel lines indicate no interaction



Non-parallel lines indicate strong interaction



# Regression Output

| Regression Statistics |              |           |         |         |       |
|-----------------------|--------------|-----------|---------|---------|-------|
| Multiple R            | 1            |           |         |         |       |
| R Square              | 1            |           |         |         |       |
| Adj R Sq              | 1            |           |         |         |       |
| Std Error             | 0.004913     |           |         |         |       |
| Observations          | 8            |           |         |         |       |
| ANOVA                 |              |           |         |         |       |
|                       | df           | SS        | MS      | F       | Sig F |
| Regression            | 3            | 492.0     | 164.0   | 6794864 | 0.000 |
| Residual              | 4            | 0.0       | 0.0     |         |       |
| Total                 | 7            | 492.0     |         |         |       |
|                       |              |           |         |         |       |
|                       | Coefficients | Std Error | t Stat  | P-value |       |
| Intercept             | 85.00        | 0.001737  | 48937.5 | 0.000   |       |
| Temp                  | -7.00        | 0.001737  | -4029.9 | 0.000   |       |
| Press                 | 3.50         | 0.001737  | 2015.4  | 0.000   |       |
| T*P                   | 0.50         | 0.001737  | 287.4   | 0.000   |       |

# Regression Output

Regression coefficients are the same as the effects determined graphically, though divided by the domain of the input, here  $2 = +1 - (-1)$ .

| Regression Statistics |              |           |         |         |       |
|-----------------------|--------------|-----------|---------|---------|-------|
| Multiple R            | 1            |           |         |         |       |
| R Square              | 1            |           |         |         |       |
| Adj R Sq              | 1            |           |         |         |       |
| Std Error             | 0.004913     |           |         |         |       |
| Observations          | 8            |           |         |         |       |
| ANOVA                 |              |           |         |         |       |
|                       | df           | SS        | MS      | F       | Sig F |
| Regression            | 3            | 492.0     | 164.0   | 6794864 | 0.000 |
| Residual              | 4            | 0.0       | 0.0     |         |       |
| Total                 | 7            | 492.0     |         |         |       |
|                       |              |           |         |         |       |
|                       | Coefficients | Std Error | t Stat  | P-value |       |
| Intercept             | 85.00        | 0.001737  | 48937.5 | 0.000   |       |
| Temp                  | -7.00        | 0.001737  | -4029.9 | 0.000   |       |
| Press                 | 3.50         | 0.001737  | 2015.4  | 0.000   |       |
| T*P                   | 0.50         | 0.001737  | 287.4   | 0.000   |       |

## EXERCISE 1.1

1. Suppose you have 4 process variables whose mean shifts you suspect contribute to the mean shift of the process output response. Of the four (A,B,C,D), you suspect there is an interact among BC but would be surprised if there are strong interactions amongst the other variables.

Design a two factorial experiment with a minimal number of runs.



# EXERCISE 1.1

There are four variables,  $n=4$ . A full factorial approach has  $2^4 = 16$  runs. 16 shutdowns of the process line. 16 days of no product being made. 16 days of zero earnings.

A half fraction design has  $2^{4-1} = 2^3 = 8$  runs. 8 shutdowns of the process line. 8 days of no product being made. 8 days of zero earnings.

A quarter fraction design has  $2^{4-2} = 2^2 = 4$  runs. 4 shutdowns of the process line. 4 days of no product being made. 4 days of zero earnings.

We would prefer as few runs as possible. On the other hand, the reason for more experiments is more explanatory terms we can explore, which has more effect, for example, A or B or C or D? We suspect BC contributes, how much compared to the others?

There are 4 main effect terms. There are  $4 \times 3 / 2 = 6$  each 2-way interaction terms. To fit up to 2-way interactions, we have  $1 + 4 + 6 = 11$  terms in the regression equation. If we don't need to disprove the significance of interaction terms not BC, then we would only have 1 interaction term (the BC term) and so we would have  $1 + 4 + 1 = 6$  terms in the regression equation.

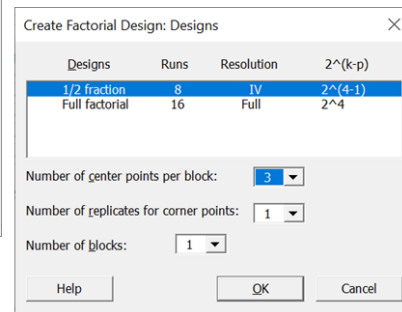
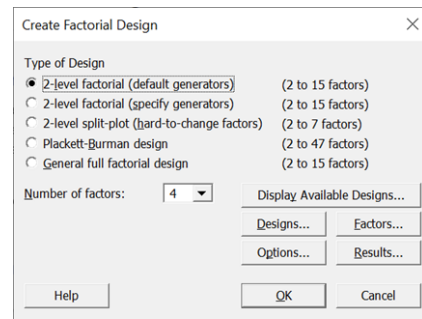
With a full factorial design, we have 16 runs with 15 degrees of freedom. We can fit up to 15 terms. This will certainly be sufficient to determine a regression equation.

With a half-fraction factorial design, we have 8 runs with 7 degrees of freedom. We can fit up to 7 terms. With a half-fraction, we can fit the main effect and BC terms, though we would not be able to disprove the other interaction terms are not significant.

With a quarter fraction design, we have 4 runs with 3 degrees of freedom. We could fit up to 3 terms. With a quarter fraction, we cannot fit the main effect terms nor BC interaction terms. We would have to make assumptions and prioritize. For example, we might do one-factor-at-a-time runs on those we think are the most likely to have an effect.

The most informative plan would be a half-fraction design. We can create this in Minitab using

Stat > DOE > Factorial Design > Create Factorial Design



# EXERCISE 1.1

In Python:

```
var_list = ["x1", "x2", "x3", "x4"]
gen = 'a b c abc'
nc = 3      # number of centerpoints
```

```
fccd = Design.from_fracfact(var_list, gen)
cccd = Design.from_centrepoints(var_list, nc)
cfd = fccd + cccd
cfd
```

```
var_list = ["x1", "x2", "x3", "x4"]
gen = 'a b c abc'
nc = 3      # number of centerpoints
|
fccd = Design.from_fracfact(var_list, gen)
cccd = Design.from_centrepoints(var_list, nc)
cfd = fccd + cccd
cfd
```

|    | PtType | Block | x1   | x2   | x3   | x4   |
|----|--------|-------|------|------|------|------|
| 1  | 1      | 1     | -1.0 | -1.0 | -1.0 | -1.0 |
| 2  | 1      | 1     | 1.0  | -1.0 | -1.0 | 1.0  |
| 3  | 1      | 1     | -1.0 | 1.0  | -1.0 | 1.0  |
| 4  | 1      | 1     | 1.0  | 1.0  | -1.0 | -1.0 |
| 5  | 1      | 1     | -1.0 | -1.0 | 1.0  | 1.0  |
| 6  | 1      | 1     | 1.0  | -1.0 | 1.0  | -1.0 |
| 7  | 1      | 1     | -1.0 | 1.0  | 1.0  | -1.0 |
| 8  | 1      | 1     | 1.0  | 1.0  | 1.0  | 1.0  |
| 9  | 0      | 1     | 0.0  | 0.0  | 0.0  | 0.0  |
| 10 | 0      | 1     | 0.0  | 0.0  | 0.0  | 0.0  |
| 11 | 0      | 1     | 0.0  | 0.0  | 0.0  | 0.0  |

## EXAMPLE 1.2

Create a full factorial DOE with 4 variables.

Consider this response, fit a model.

Complete a residual analysis.

| RUN | STD | BLOCK | CP | A | B  | C  | D  | RESPONSE |
|-----|-----|-------|----|---|----|----|----|----------|
| 1   | 1   | 1     | 1  | 1 | -1 | -1 | -1 | 6.899    |
| 2   | 2   | 2     | 1  | 1 | 1  | -1 | -1 | 9.769    |
| 3   | 3   | 3     | 1  | 1 | -1 | 1  | -1 | 7.931    |
| 4   | 4   | 4     | 1  | 1 | 1  | 1  | -1 | 8.038    |
| 5   | 5   | 5     | 1  | 1 | -1 | -1 | 1  | 11.366   |
| 6   | 6   | 6     | 1  | 1 | 1  | -1 | 1  | 13.824   |
| 7   | 7   | 7     | 1  | 1 | -1 | 1  | 1  | 11.540   |
| 8   | 8   | 8     | 1  | 1 | 1  | 1  | 1  | 11.659   |
| 9   | 9   | 9     | 1  | 1 | -1 | -1 | -1 | 7.580    |
| 10  | 10  | 10    | 1  | 1 | 1  | -1 | -1 | 9.845    |
| 11  | 11  | 11    | 1  | 1 | -1 | 1  | -1 | 7.008    |
| 12  | 12  | 12    | 1  | 1 | 1  | 1  | -1 | 7.753    |
| 13  | 13  | 13    | 1  | 1 | -1 | -1 | 1  | 10.296   |
| 14  | 14  | 14    | 1  | 1 | 1  | -1 | 1  | 12.306   |
| 15  | 15  | 15    | 1  | 1 | -1 | 1  | 1  | 10.783   |
| 16  | 16  | 16    | 1  | 1 | 1  | 1  | 1  | 9.923    |

# EXAMPLE 1.2

Define Custom Factorial Design

Factors:

A-D

2-level factorial  
2-level split-plot (hard-to-change)  
General full factorial

Factors from above that are hard to change

Select

Low/High

Help

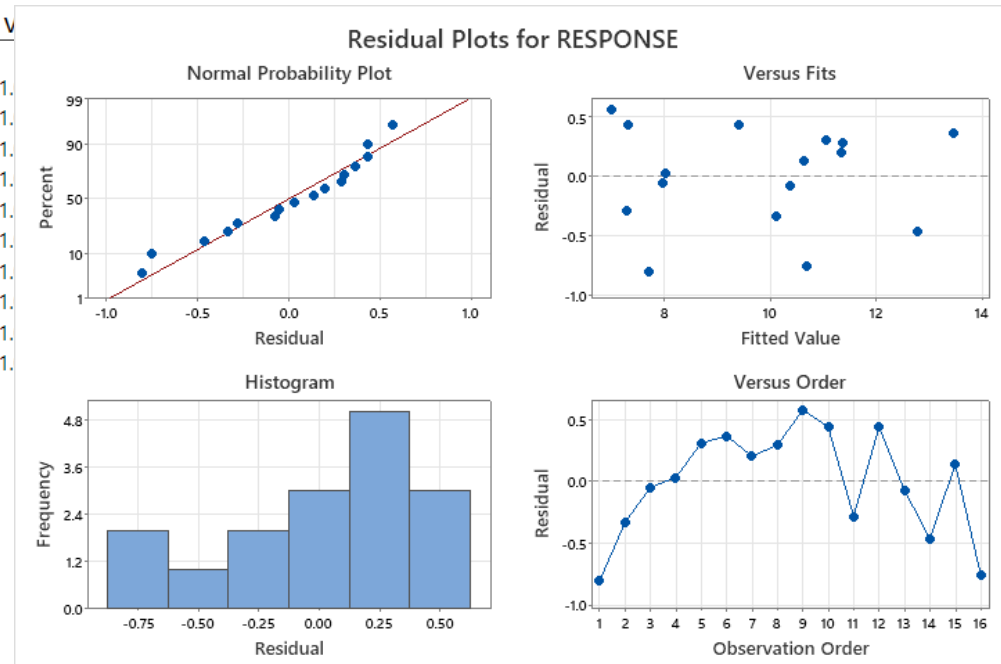
OK

## Coded Coefficients

| Term     | Effect  | Coef    | SE Coef | T-Value | P-Value | V  |
|----------|---------|---------|---------|---------|---------|----|
| Constant |         | 9.7825  | 0.0922  | 106.06  | 0.000   |    |
| A        | 1.2142  | 0.6071  | 0.0922  | 6.58    | 0.001   | 1. |
| B        | -0.9063 | -0.4532 | 0.0922  | -4.91   | 0.004   | 1. |
| C        | 3.3592  | 1.6796  | 0.0922  | 18.21   | 0.000   | 1. |
| D        | -0.6916 | -0.3458 | 0.0922  | -3.75   | 0.013   | 1. |
| A*B      | -1.1863 | -0.5931 | 0.0922  | -6.43   | 0.001   | 1. |
| A*C      | -0.2820 | -0.1410 | 0.0922  | -1.53   | 0.187   | 1. |
| A*D      | -0.1740 | -0.0870 | 0.0922  | -0.94   | 0.389   | 1. |
| B*C      | -0.0655 | -0.0327 | 0.0922  | -0.35   | 0.737   | 1. |
| B*D      | -0.2337 | -0.1169 | 0.0922  | -1.27   | 0.261   | 1. |
| C*D      | -0.5787 | -0.2894 | 0.0922  | -3.14   | 0.026   | 1. |

## Model Summary

| S        | R-sq   | R-sq(adj) | R-sq(pred) |
|----------|--------|-----------|------------|
| 0.368929 | 98.95% | 96.84%    | 89.21%     |



# EXAMPLE 1.2

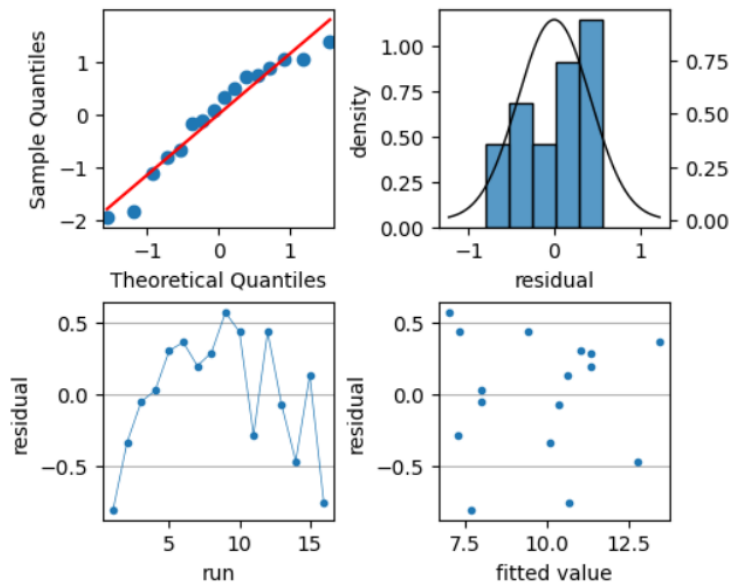
```
var_list = ["x1", "x2", "x3", "x4"]
levels = [2, 2, 2, 2]
ffd = Design.from_fullfact(var_list, levels)
df_ffd = ffd.to_df()
df_ffd['Response'] = [6.899, 9.769, 7.931, 8.038, 11.366, 13.824,
                      11.540, 11.659, 7.580, 9.845, 7.008, 7.753,
                      10.296, 12.306, 10.783, 9.923]
```

```
#form = 'Response ~ x1 + x2 + x3 + x4 + x1*x2 + x1*x3 + x1*x4 + x2*x3 + x2*x4 + x3*x4'
form = 'Response ~ x1 + x2 + x3 + x4 + x1*x2'
mod = smf.ols(form, df_ffd)
res = mod.fit()
```

```
with Figure(5, 4, 2, 2) as (fig, ax):
    mqr.plot_regression_residuals(res, axs=ax)
    plot = grab_figure(fig)
```

```
hstack(
    vstack(
        mqr.anova.adequacy(res),
        mqr.anova.coeffs(res)
    ),
    plot
)
```

| S         | R-sq      | R-sq (adj) | N         | JB           | p (JB)   |
|-----------|-----------|------------|-----------|--------------|----------|
| 0.51929   | 0.958232  | 0.937348   | 16        | 1.296005     | 0.523089 |
|           | Coeff     | [2.5%      | 97.5%]    | PR(> F)      | VIF      |
| Intercept | 9.782500  | 9.493238   | 10.071762 | 4.136163e-15 | 1.0      |
| x1        | 0.607125  | 0.317863   | 0.896387  | 8.720376e-04 | 1.0      |
| x2        | -0.453125 | -0.742387  | -0.163863 | 5.819665e-03 | 1.0      |
| x3        | 1.679625  | 1.390363   | 1.968887  | 1.435277e-07 | 1.0      |
| x4        | -0.345750 | -0.635012  | -0.056488 | 2.376622e-02 | 1.0      |
| x1:x2     | -0.593250 | -0.882512  | -0.303988 | 1.026728e-03 | 1.0      |





# Summary

## Experimentation

- Clarify a measurable response
- Clarify possible factors to change
  - Look at root causes. Ask why? Ask again, why the why? And again.
- Graph the results

## Design of experiments is for multiple inputs

- Orthogonal arrays chosen to investigate particular terms
- Regression statistical analysis to determine a response equation
- Scrutinize results for statistical significance



# Overview

Experimentation

Effects and Interactions

Full Factorial Experiment

Fractional Factorial Designs

Center Points

Central Composite Designs

Computer Experiments and Latin Hypercubes



# Full Factorial Models

Suppose we have 10 design variables. 2-level experiments.

$2^{10} = 1024$  experimental points!

How many coefficients could we fit?

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{12} x_1 x_2 + \cdots + \beta_{123} x_1 x_2 x_3 + \cdots$$



# Full Factorial Inefficiencies

| # effects | Effect Type        | Term             |
|-----------|--------------------|------------------|
| 1         | Average response   | $\beta_0$        |
| 10        | Main effects       | $\beta_i$        |
| 45        | 2-way interactions | $\beta_{ij}$     |
| 120       | 3-way interactions | $\beta_{ijk}$    |
| 210       | 4-way interactions | $\beta_{ijkl}$   |
| 252       | 5-way interactions | $\beta_{ijklm}$  |
| 210       | 6-way interactions | $\beta_{ij...}$  |
| 120       | 7-way interactions | $\beta_{ij...}$  |
| 45        | 8-way interactions | $\beta_{ij...}$  |
| 10        | 9-way interactions | $\beta_{ij...}$  |
| 1         | 10-way interaction | $\beta_{ij...n}$ |
| 1024      | TOTAL              |                  |

# Reduce the number

For a complete 2-way interaction model

- $m_i$  is the number of levels experimented on  $x_i$

$$N = 1 + \sum_{i=1}^n (m_i - 1) + \sum_{i=1}^n \sum_{j=i+1}^n (m_i - 1)(m_j - 1)$$

- For 2 level experiments

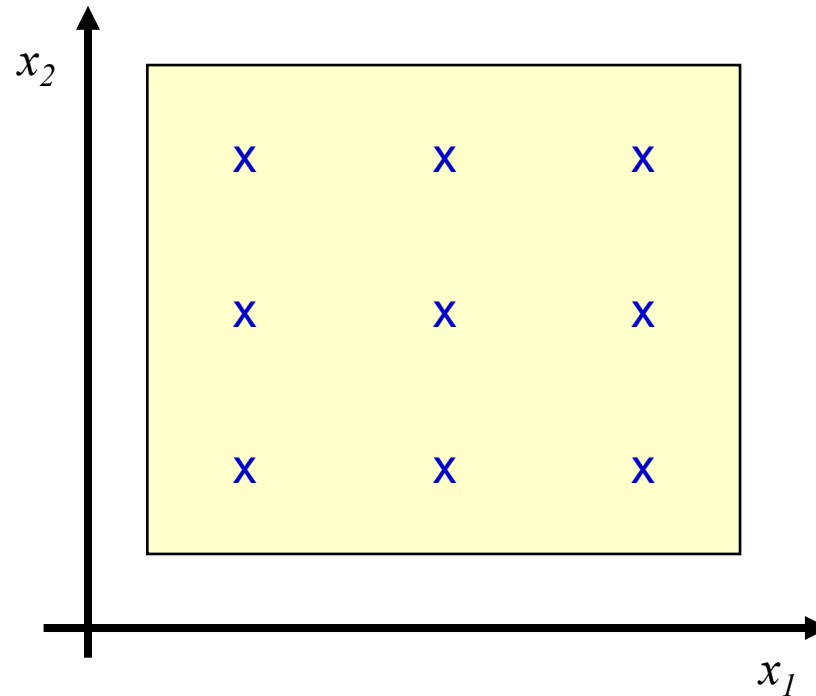
$$N = 1 + n + \frac{n(n-1)}{2}$$

We need to find a set of  $N$  experimental points  $\vec{x}$  that can fit each coefficient.

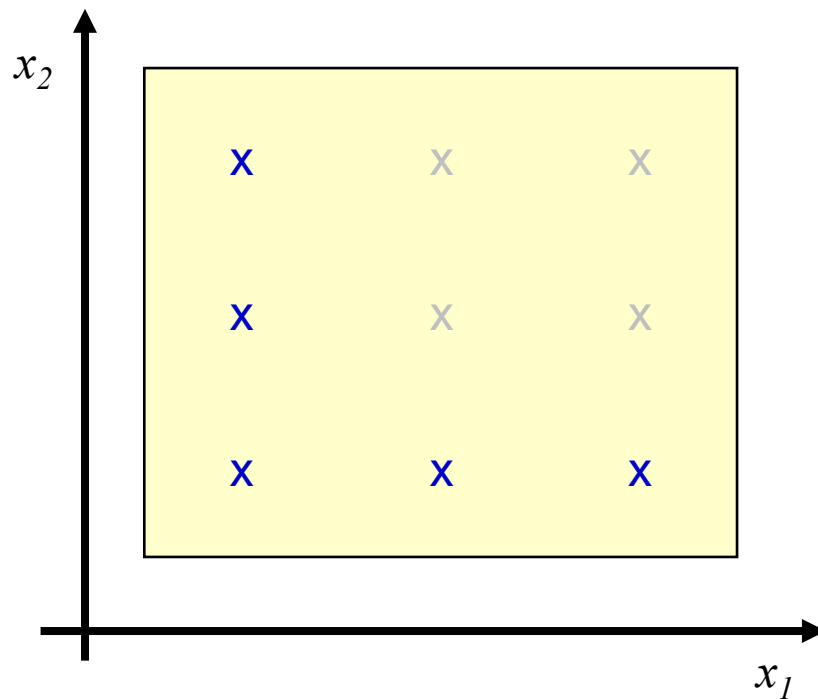
# Reducing the space

A  $2^3$  design has 9 runs. Suppose you only want to do 5.

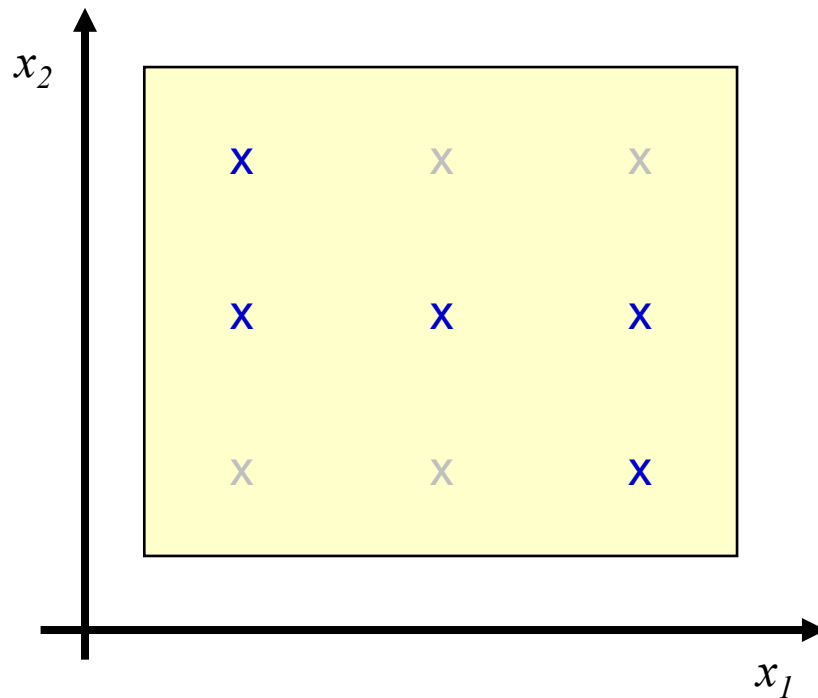
Which 5?



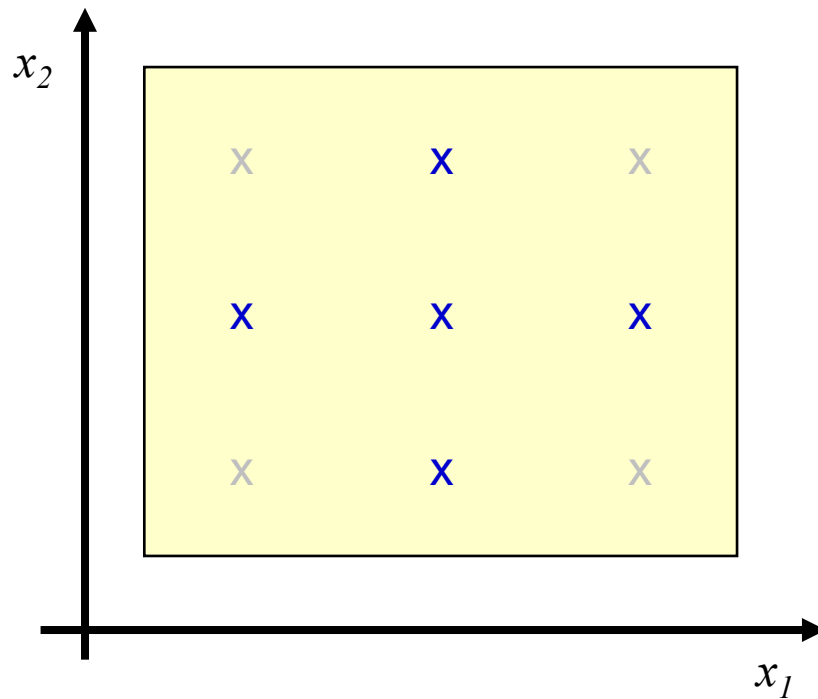
# Reducing the space



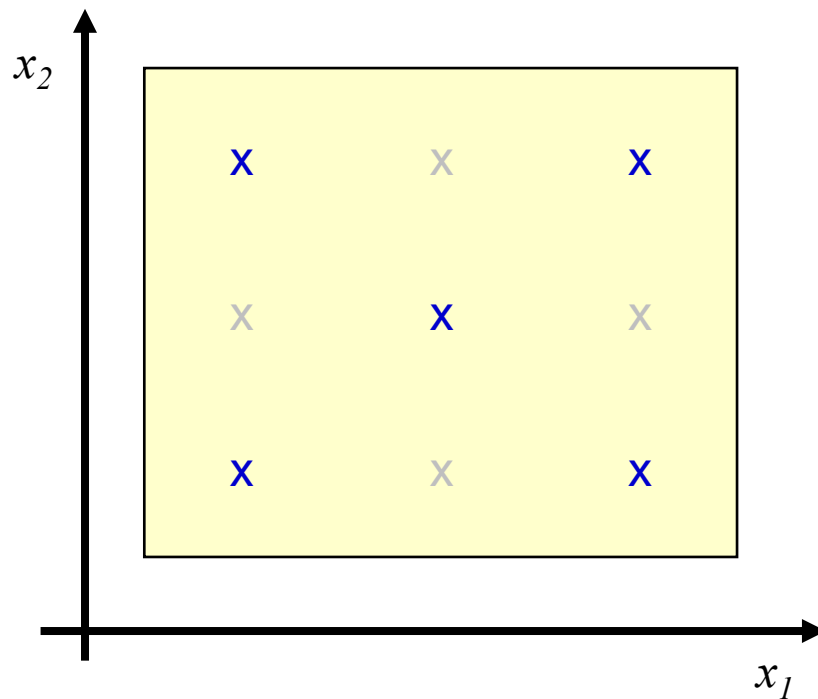
# Reducing the space



# Reducing the space



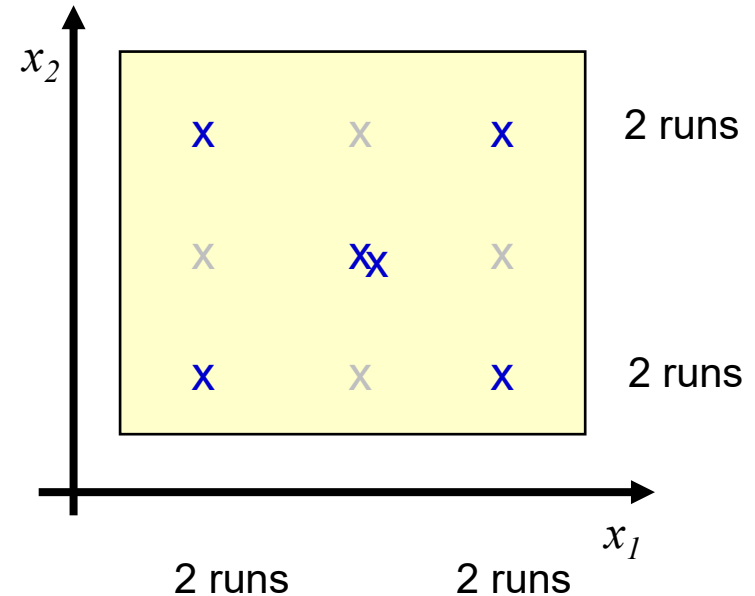
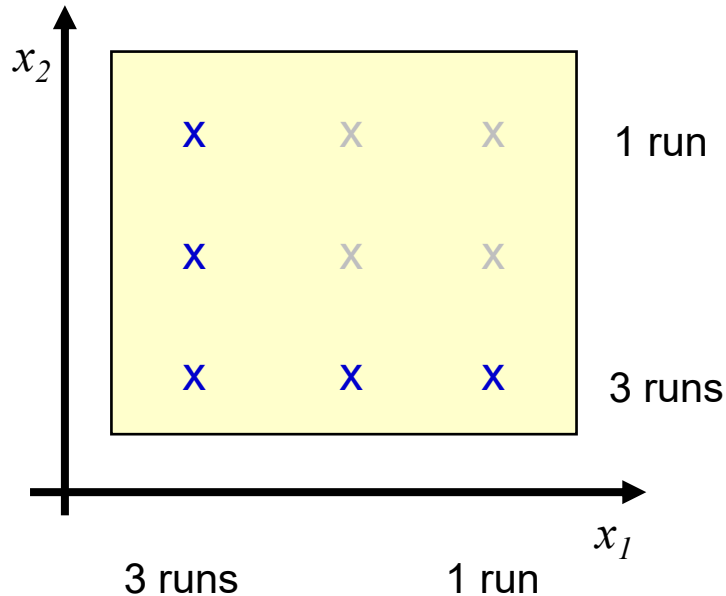
# Reducing the space



# Orthogonal and Balanced

Want the reduced set of experiments to “cover” the space

- This means an equal number of runs at each level  $i$  for all  $x_i$ .
  - Except at the center point.







# Orthogonal and Balanced

Linear algebra: “well cover” means orthogonal and balanced

With coded units, need all  $\bar{x}_i$  and  $\bar{x}_j$  such that

$$\sum \bar{x}_i \bar{x}_j = 0$$

These designs have been pre-determined over the years.

- Two factorial designs –  $2^{k-p}$  designs
- N-factorial designs – specialized, beyond this course.



# Yates Labeling

Relabel each  $x_i$  alphabetically

$$x_1 = a$$

$$x_2 = b$$

$$x_3 = c$$

Relabel interactions alphabetically

$$x_1x_2 = ab$$

$$x_3x_4 = cd$$

$$x_1x_2x_4 = acd$$

# Yates Order and Interactions

Consider the  $2^6$  design matrix

| Run | Design Variable (Yates Order) |         |         |         |         |         |
|-----|-------------------------------|---------|---------|---------|---------|---------|
|     | $x_1=a$                       | $x_2=b$ | $x_3=c$ | $x_4=d$ | $x_5=e$ | $x_6=f$ |
| 1   | -1                            | -1      | -1      | -1      | -1      | -1      |
| 2   | 1                             | -1      | -1      | -1      | -1      | -1      |
| 3   | -1                            | 1       | -1      | -1      | -1      | -1      |
| 4   | 1                             | 1       | -1      | -1      | -1      | -1      |
| 5   | -1                            | -1      | 1       | -1      | -1      | -1      |
| 6   | 1                             | -1      | 1       | -1      | -1      | -1      |
| 7   | -1                            | 1       | 1       | -1      | -1      | -1      |
| 8   | 1                             | 1       | 1       | -1      | -1      | -1      |
| 9   | -1                            | -1      | -1      | 1       | -1      | -1      |
| 10  | 1                             | -1      | -1      | 1       | -1      | -1      |
| :   |                               |         |         |         |         |         |
| 63  | -1                            | 1       | 1       | 1       | 1       | 1       |
| 64  | 1                             | 1       | 1       | 1       | 1       | 1       |

# Yates Order and Interactions

Examine the design matrix interactions

| Run | Design Variable (Yates Order) |         |         |         |         |         |    |     |        |
|-----|-------------------------------|---------|---------|---------|---------|---------|----|-----|--------|
|     | $x_1=a$                       | $x_2=b$ | $x_3=c$ | $x_4=d$ | $x_5=e$ | $x_6=f$ | ab | acd | abcdef |
| 1   | -1                            | -1      | -1      | -1      | -1      | -1      | 1  | -1  | 1      |
| 2   | 1                             | -1      | -1      | -1      | -1      | -1      | -1 | 1   | -1     |
| 3   | -1                            | 1       | -1      | -1      | -1      | -1      | -1 | -1  | -1     |
| 4   | 1                             | 1       | -1      | -1      | -1      | -1      | 1  | 1   | 1      |
| 5   | -1                            | -1      | 1       | -1      | -1      | -1      | 1  | 1   | -1     |
| 6   | 1                             | -1      | 1       | -1      | -1      | -1      | -1 | -1  | 1      |
| 7   | -1                            | 1       | 1       | -1      | -1      | -1      | -1 | 1   | 1      |
| 8   | 1                             | 1       | 1       | -1      | -1      | -1      | 1  | -1  | -1     |
| 9   | -1                            | -1      | -1      | 1       | -1      | -1      | 1  | 1   | -1     |
| 10  | 1                             | -1      | -1      | 1       | -1      | -1      | -1 | -1  | 1      |
| :   |                               |         |         |         |         |         |    |     |        |
| 63  | -1                            | 1       | 1       | 1       | 1       | 1       | -1 | -1  | -1     |
| 64  | 1                             | 1       | 1       | 1       | 1       | 1       | 1  | 1   | 1      |

# Blocks

Consider any interaction. Partition the matrix according to it. Each half defines a **block**.

| Run | Design Variable (Yates Order) |         |         |         |         |         | abcdef | BLOCK |
|-----|-------------------------------|---------|---------|---------|---------|---------|--------|-------|
|     | $x_1=a$                       | $x_2=b$ | $x_3=c$ | $x_4=d$ | $x_5=e$ | $x_6=f$ |        |       |
| 1   | -1                            | -1      | -1      | -1      | -1      | -1      | 1      | I     |
| 2   | 1                             | -1      | -1      | -1      | -1      | -1      | -1     | II    |
| 3   | -1                            | 1       | -1      | -1      | -1      | -1      | -1     | II    |
| 4   | 1                             | 1       | -1      | -1      | -1      | -1      | 1      | I     |
| 5   | -1                            | -1      | 1       | -1      | -1      | -1      | -1     | II    |
| 6   | 1                             | -1      | 1       | -1      | -1      | -1      | 1      | I     |
| 7   | -1                            | 1       | 1       | -1      | -1      | -1      | 1      | I     |
| 8   | 1                             | 1       | 1       | -1      | -1      | -1      | -1     | II    |
| 9   | -1                            | -1      | -1      | 1       | -1      | -1      | -1     | II    |
| 10  | 1                             | -1      | -1      | 1       | -1      | -1      | 1      | I     |
| :   |                               |         |         |         |         |         |        |       |
| 63  | -1                            | 1       | 1       | 1       | 1       | 1       | -1     | II    |
| 64  | 1                             | 1       | 1       | 1       | 1       | 1       | 1      | I     |

# Blocking

Each block has  $1/2$  as many runs. Each block **confounds** the **generator** of the block.

| Run | Design Variable (Yates Order) |         |         |         |         |         | abcdef | BLOCK |
|-----|-------------------------------|---------|---------|---------|---------|---------|--------|-------|
|     | $x_1=a$                       | $x_2=b$ | $x_3=c$ | $x_4=d$ | $x_5=e$ | $x_6=f$ |        |       |
| 1   | -1                            | -1      | -1      | -1      | -1      | -1      | 1      | I     |
| 2   | 1                             | -1      | -1      | -1      | -1      | -1      | -1     | II    |
| 3   | -1                            | 1       | -1      | -1      | -1      | -1      | -1     | II    |
| 4   | 1                             | 1       | -1      | -1      | -1      | -1      | 1      | I     |
| 5   | -1                            | -1      | 1       | -1      | -1      | -1      | -1     | II    |
| 6   | 1                             | -1      | 1       | -1      | -1      | -1      | 1      | I     |
| 7   | -1                            | 1       | 1       | -1      | -1      | -1      | 1      | I     |
| 8   | 1                             | 1       | 1       | -1      | -1      | -1      | -1     | II    |
| 9   | -1                            | -1      | -1      | 1       | -1      | -1      | -1     | II    |
| 10  | 1                             | -1      | -1      | 1       | -1      | -1      | 1      | I     |
| :   |                               |         |         |         |         |         |        |       |
| 63  | -1                            | 1       | 1       | 1       | 1       | 1       | -1     | II    |
| 64  | 1                             | 1       | 1       | 1       | 1       | 1       | 1      | I     |

abcdef = +1

abcdef = -1

# Blocking

Each block has 1/2 as many experiments. **32 runs instead of 64.**

However, each block *confounds* the *generator* of the block.

| Run | Design Variable (Yates Order) |         |         |         |         |         | abcdef | BLOCK |
|-----|-------------------------------|---------|---------|---------|---------|---------|--------|-------|
|     | $x_1=a$                       | $x_2=b$ | $x_3=c$ | $x_4=d$ | $x_5=e$ | $x_6=f$ |        |       |
| 1   | -1                            | -1      | -1      | -1      | -1      | -1      | 1      | I     |
| 4   | 1                             | 1       | -1      | -1      | -1      | -1      | 1      | I     |
| 6   | 1                             | -1      | 1       | -1      | -1      | -1      | 1      | I     |
| 7   | -1                            | 1       | 1       | -1      | -1      | -1      | 1      | I     |
| 10  | 1                             | -1      | -1      | 1       | -1      | -1      | 1      | I     |
| :   |                               |         |         |         |         |         |        |       |
| 64  | 1                             | 1       | 1       | 1       | 1       | 1       | 1      | I     |

abcdef = +1

# Further interactions

Consider the  $2^{4-1}$  block with  $abcd = +1$

| Exp. | $x_1$<br>a | $x_2$<br>b | $x_3$<br>c | $x_4$<br>d=abc | ab | ac | ad | bc | bd | cd | abc | abd | acd | bcd | abcd | 1 |
|------|------------|------------|------------|----------------|----|----|----|----|----|----|-----|-----|-----|-----|------|---|
| 1    | -1         | -1         | -1         | -1             | 1  | 1  | 1  | 1  | 1  | 1  | -1  | -1  | -1  | -1  | 1    | 1 |
| 2    | 1          | -1         | -1         | 1              | -1 | -1 | 1  | 1  | -1 | -1 | 1   | -1  | -1  | 1   | 1    | 1 |
| 3    | -1         | 1          | -1         | 1              | -1 | 1  | -1 | -1 | 1  | -1 | 1   | -1  | 1   | -1  | 1    | 1 |
| 4    | 1          | 1          | -1         | -1             | 1  | -1 | -1 | -1 | -1 | 1  | -1  | -1  | 1   | 1   | 1    | 1 |
| 5    | -1         | -1         | 1          | 1              | 1  | -1 | -1 | -1 | -1 | 1  | 1   | 1   | -1  | -1  | 1    | 1 |
| 6    | 1          | -1         | 1          | -1             | -1 | 1  | -1 | -1 | 1  | -1 | -1  | 1   | -1  | 1   | 1    | 1 |
| 7    | -1         | 1          | 1          | -1             | -1 | -1 | 1  | 1  | -1 | -1 | -1  | 1   | 1   | -1  | 1    | 1 |
| 8    | 1          | 1          | 1          | 1              | 1  | 1  | 1  | 1  | 1  | 1  | 1   | 1   | 1   | 1   | 1    | 1 |



# Further interactions

Consider the  $2^{4-1}$  block with  $abcd = +1$

| Exp. | $x_1$<br>a | $x_2$<br>b | $x_3$<br>c | $x_4$<br>d=abc | ab | ac | ad | bc | bd | cd | abc | abd | acd | bcd | abcd | 1 |
|------|------------|------------|------------|----------------|----|----|----|----|----|----|-----|-----|-----|-----|------|---|
| 1    | -1         | -1         | -1         | -1             | 1  | 1  | 1  | 1  | 1  | 1  | -1  | -1  | -1  | -1  | 1    | 1 |
| 2    | 1          | -1         | -1         | 1              | -1 | -1 | 1  | 1  | -1 | -1 | 1   | -1  | -1  | 1   | 1    | 1 |
| 3    | -1         | 1          | -1         | 1              | -1 | 1  | -1 | -1 | 1  | -1 | 1   | -1  | 1   | -1  | 1    | 1 |
| 4    | 1          | 1          | -1         | -1             | 1  | -1 | -1 | -1 | -1 | 1  | -1  | -1  | 1   | 1   | 1    | 1 |
| 5    | -1         | -1         | 1          | 1              | 1  | -1 | -1 | -1 | -1 | 1  | 1   | 1   | -1  | -1  | 1    | 1 |
| 6    | 1          | -1         | 1          | -1             | -1 | 1  | -1 | -1 | 1  | -1 | -1  | 1   | -1  | 1   | 1    | 1 |
| 7    | -1         | 1          | 1          | -1             | -1 | -1 | 1  | 1  | -1 | -1 | -1  | 1   | 1   | -1  | 1    | 1 |
| 8    | 1          | 1          | 1          | 1              | 1  | 1  | 1  | 1  | 1  | 1  | 1   | 1   | 1   | 1   | 1    | 1 |

Compare a with bcd, b with acd, c with abd and d with abc.

# Further interactions

Consider the  $2^{4-1}$  block with  $abcd = +1$

| Exp. | $x_1$<br>a | $x_2$<br>b | $x_3$<br>c | $x_4$<br>d=abc | ab | ac | ad | bc | bd | cd | abc | abd | acd | bcd | abcd | 1 |
|------|------------|------------|------------|----------------|----|----|----|----|----|----|-----|-----|-----|-----|------|---|
| 1    | -1         | -1         | -1         | -1             | 1  | 1  | 1  | 1  | 1  | 1  | -1  | -1  | -1  | -1  | 1    | 1 |
| 2    | 1          | -1         | -1         | 1              | -1 | -1 | 1  | 1  | -1 | -1 | 1   | -1  | -1  | 1   | 1    | 1 |
| 3    | -1         | 1          | -1         | 1              | -1 | 1  | -1 | -1 | 1  | -1 | 1   | -1  | 1   | -1  | 1    | 1 |
| 4    | 1          | 1          | -1         | -1             | 1  | -1 | -1 | -1 | -1 | 1  | -1  | -1  | 1   | 1   | 1    | 1 |
| 5    | -1         | -1         | 1          | 1              | 1  | -1 | -1 | -1 | -1 | 1  | 1   | 1   | -1  | -1  | 1    | 1 |
| 6    | 1          | -1         | 1          | -1             | -1 | 1  | -1 | -1 | 1  | -1 | -1  | 1   | -1  | 1   | 1    | 1 |
| 7    | -1         | 1          | 1          | -1             | -1 | -1 | 1  | 1  | -1 | -1 | -1  | 1   | 1   | -1  | 1    | 1 |
| 8    | 1          | 1          | 1          | 1              | 1  | 1  | 1  | 1  | 1  | 1  | 1   | 1   | 1   | 1   | 1    | 1 |

Compare a with bcd, b with acd, c with abd and d with abc.

These terms are **confounded**. If using this matrix results in 'a' as significant, it might be actually be 'bcd.'

# Confounded Variables

$$\begin{array}{cccc} a = bcd & ab = cd & b = acd & ac = bd \\ c = abd & bc = ad & d = abc & 1 = abcd \end{array}$$

This happens with partial factorial designs. More generally, a fraction factorial design has partial confounding, or *aliasing*.

$$a = 1/3 bc + 1/3 bd + 1/3 cd$$

The trick is to select a good block for the phenomena you want to study.

- Where the confounded interaction is one you don't care about.

# Design Resolution

The *design resolution* is a rating of the level of interactions a design matrix can fit without confounding

|                |                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resolution III | No Main effect is confounded with another. Main effects and 2-factor interactions are confounded with each other.                                                                                |
| Resolution IV  | Main Effects are not confounded with each other or with 2-factor interactions. Main effects are confounded with 3-factor interactions, and 2-factor interactions are confounded with each other. |
| Resolution V   | No Main Effect or 2-factor interaction effect is confounded with another. Main Effects are confounded with 4-Factor interactions, 2-Factor interactions with 3-Factors.                          |



# Design Resolutions

The resolution of a 2-level fractional factorial design is the smallest sum of the orders of aliased effects.

| <u>Resolution</u> | <u>Smallest Sum of Aliased Effects</u>      |
|-------------------|---------------------------------------------|
| III               | 1+2 (main effect with 2-Factor Interaction) |
| IV                | 1+3 , 2+2                                   |
| V                 | 1+4 , 2+3                                   |

The resolution of a fractional factorial design is often included as a subscript in the designation of the design, e.g. a  $2^{4-1}$  Res IV design.

# Design Resolution

|                           |     | Number of Factors |      |      |      |      |      |     |     |     |     |     |     |     |     |
|---------------------------|-----|-------------------|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-----|
| Treatment<br>Combinations |     | 2                 | 3    | 4    | 5    | 6    | 7    | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  |
|                           | 4   | Full              | III  |      |      |      |      |     |     |     |     |     |     |     |     |
|                           | 8   |                   | Full | IV   | III  | III  | III  |     |     |     |     |     |     |     |     |
|                           | 16  |                   |      | Full | V    | IV   | IV   | IV  | III | III | III | III | III | III | III |
|                           | 32  |                   |      |      | Full | VI   | IV   | IV  | IV  | IV  | IV  | IV  | IV  | IV  | IV  |
|                           | 64  |                   |      |      |      | Full | VII  | V   | IV  | IV  | IV  | IV  | IV  | IV  | IV  |
|                           | 128 |                   |      |      |      |      | Full | VII | VI  | V   | V   | IV  | IV  | IV  | IV  |

# Fractional Factorials – Rules

Rule #1. Do not confound main effects with each other.

- When can you violate this rule?

Rule #2. Do not confound main effects with two factor interactions.

- When can you violate this rule?

Rule #3. Do not confound two factor interactions with each other.

- When can you violate this rule?



# Two Factorial Designs – Minitab

To create factorial designs in Minitab

Stat > DOE > Factorial > Create Factorial Design...

Create Factorial Design

Type of Design

- ☒ 2-level factorial (default generators) (2 to 15 factors)
- ☐ 2-level factorial (specify generators) (2 to 15 factors)
- ☐ 2-level split-plot (hard-to-change factors) (2 to 7 factors)
- ☐ Plackett-Burman design (2 to 47 factors)
- ☐ General full factorial design (2 to 15 factors)

Number of factors:

Display Available Designs...

Designs... Factors...

Options... Results...

Help OK Cancel

Create Factorial Design: Designs

| Designs        | Runs | Resolution | $2^{(k-p)}$ |
|----------------|------|------------|-------------|
| 1/2 fraction   | 8    | IV         | $2^{(4-1)}$ |
| Full factorial | 16   | Full       | $2^4$       |

Number of center points per block:

Number of replicates for corner points:

Number of blocks:

Help OK Cancel





# Two Factorial Designs – Python

$2^{n-k}$  fractional factorial arrays are easily created in Python

```
X = fracfact(string)
```

- where string is a text string of the generator such as 'a b c d bcd acd'
- The result is in 'coded units' (-1, 1)
- Create a second 'uncoded units' matrix replacing these with the levels of each variable

```
>> from pyDOE2 import *
>> X = fracfact('a b c d bcd acd')

X =
-1 -1 -1 -1 -1 -1
-1 -1 -1 1 1 1
-1 -1 1 -1 1 1
-1 -1 1 1 -1 -1
-1 1 -1 -1 1 -1
-1 1 -1 1 -1 1
-1 1 1 -1 -1 1
-1 1 1 1 1 -1
1 -1 -1 -1 -1 1
1 -1 -1 1 1 -1
1 -1 1 -1 1 -1
1 -1 1 1 -1 1
1 1 -1 -1 1 1
1 1 -1 1 -1 -1
1 1 1 -1 -1 -1
1 1 1 1 1 1
0 0 0
```

# Standard DOE Arrays

Available standard DOE arrays are available

L4, L8, L16, etc

| L4                                               | x <sub>1</sub> | x <sub>2</sub> | x <sub>3</sub> |
|--------------------------------------------------|----------------|----------------|----------------|
|                                                  | A              | B              | C              |
| 1                                                | -1             | -1             | 1              |
| 2                                                | 1              | -1             | -1             |
| 3                                                | -1             | 1              | -1             |
| 4                                                | 1              | 1              | 1              |
| Main Effect<br>Confounded<br>with<br>Interaction | BC             | AC             | AB             |

| L8                                               | x <sub>1</sub> | x <sub>2</sub> | x <sub>3</sub> | x <sub>4</sub> | x <sub>5</sub> | x <sub>6</sub> | x <sub>7</sub> |
|--------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                                                  | A              | B              | C              | D              | E              | F              | G              |
| 1                                                | -1             | -1             | -1             | -1             | 1              | 1              | 1              |
| 2                                                | 1              | -1             | -1             | 1              | 1              | -1             | -1             |
| 3                                                | -1             | 1              | -1             | 1              | -1             | 1              | -1             |
| 4                                                | 1              | 1              | -1             | -1             | -1             | -1             | 1              |
| 5                                                | -1             | -1             | 1              | 1              | -1             | -1             | 1              |
| 6                                                | 1              | -1             | 1              | -1             | -1             | 1              | -1             |
| 7                                                | -1             | 1              | 1              | -1             | 1              | -1             | -1             |
| 8                                                | 1              | 1              | 1              | 1              | 1              | 1              | 1              |
| Main Effect<br>Confounded<br>with<br>Interaction | DE             | DF             | DG             | CG             | FG             | EG             | EF             |
|                                                  | CF             | CE             | BE             | BF             | BC             | BD             | CD             |
|                                                  | BG             | AG             | AF             | AE             | AD             | AC             | AB             |

# DOE Arrays

Available standard DOE arrays are available

L4, L8, L16, etc

| L16                                              | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | X <sub>6</sub> | X <sub>7</sub> | X <sub>8</sub> | X <sub>9</sub> | X <sub>10</sub> | X <sub>11</sub> | X <sub>12</sub> | X <sub>13</sub> | X <sub>14</sub> | X <sub>15</sub> |
|--------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|                                                  | A              | B              | C              | D              | E              | F              | G              | H              | I              | J               | K               | L               | M               | N               | O               |
| 1                                                | -1             | -1             | -1             | -1             | 1              | -1             | -1             | -1             | -1             | 1               | 1               | 1               | 1               | 1               | 1               |
| 2                                                | 1              | -1             | -1             | -1             | -1             | 1              | 1              | 1              | -1             | -1              | -1              | -1              | 1               | 1               | 1               |
| 3                                                | -1             | 1              | -1             | -1             | -1             | 1              | 1              | -1             | 1              | -1              | 1               | 1               | -1              | -1              | 1               |
| 4                                                | 1              | 1              | -1             | -1             | 1              | -1             | -1             | 1              | 1              | 1               | -1              | -1              | -1              | -1              | 1               |
| 5                                                | -1             | -1             | 1              | -1             | -1             | 1              | -1             | 1              | 1              | 1               | -1              | 1               | -1              | 1               | -1              |
| 6                                                | 1              | -1             | 1              | -1             | 1              | -1             | 1              | -1             | 1              | -1              | 1               | -1              | -1              | 1               | -1              |
| 7                                                | -1             | 1              | 1              | -1             | 1              | -1             | 1              | 1              | -1             | -1              | -1              | 1               | 1               | -1              | -1              |
| 8                                                | 1              | 1              | 1              | -1             | -1             | 1              | -1             | -1             | -1             | 1               | 1               | -1              | 1               | -1              | -1              |
| 9                                                | -1             | -1             | -1             | 1              | -1             | -1             | 1              | 1              | 1              | 1               | 1               | -1              | 1               | -1              | -1              |
| 10                                               | 1              | -1             | -1             | 1              | 1              | 1              | -1             | -1             | 1              | -1              | -1              | 1               | 1               | -1              | -1              |
| 11                                               | -1             | 1              | -1             | 1              | 1              | 1              | -1             | 1              | -1             | -1              | 1               | -1              | -1              | 1               | -1              |
| 12                                               | 1              | 1              | -1             | 1              | -1             | -1             | 1              | -1             | -1             | 1               | -1              | 1               | -1              | 1               | -1              |
| 13                                               | -1             | -1             | 1              | 1              | 1              | 1              | 1              | -1             | -1             | 1               | -1              | -1              | -1              | -1              | 1               |
| 14                                               | 1              | -1             | 1              | 1              | -1             | -1             | -1             | 1              | -1             | -1              | 1               | 1               | -1              | -1              | 1               |
| 15                                               | -1             | 1              | 1              | 1              | -1             | -1             | -1             | -1             | 1              | -1              | -1              | -1              | 1               | 1               | 1               |
| 16                                               | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1               | 1               | 1               | 1               | 1               | 1               |
| Main Effect<br>Confounded<br>with<br>Interaction | BJ             | AJ             | AK             | AL             | AI             | AM             | AN             | AO             | AE             | AB              | AC              | AD              | AF              | AG              | AH              |
|                                                  | CK             | CM             | BM             | BN             | BH             | BK             | BL             | BE             | BO             | CF              | BF              | BG              | BC              | BD              | BI              |
|                                                  | DL             | DN             | DO             | CO             | CG             | CJ             | CE             | CL             | CN             | DG              | DH              | CH              | DI              | CI              | CD              |
|                                                  | EI             | EH             | EG             | EF             | DF             | DE             | DJ             | DK             | DM             | EO              | EN              | EM              | EL              | EK              | EJ              |
|                                                  | FM             | FK             | FJ             | GJ             | JO             | GO             | FO             | FN             | FL             | HI              | GI              | FI              | GH              | HF              | FG              |
|                                                  | GN             | GL             | HL             | HK             | KN             | HN             | HM             | GM             | GK             | KM              | JM              | JN              | JK              | JL              | KL              |
|                                                  | HO             | IO             | IN             | IM             | LM             | IL             | IK             | JI             | HJ             | LN              | LO              | OK              | ON              | OM              | MN              |

# EXAMPLE

Take the previous full factorial DOE and make a  $\frac{1}{2}$  fraction of it

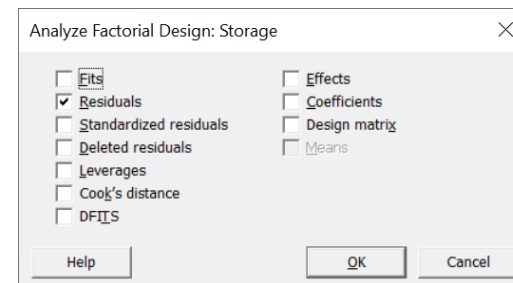
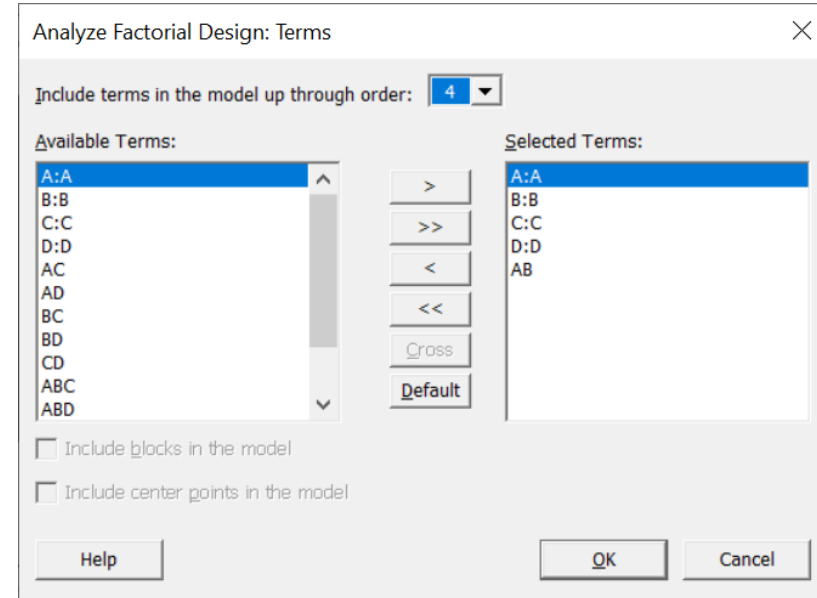
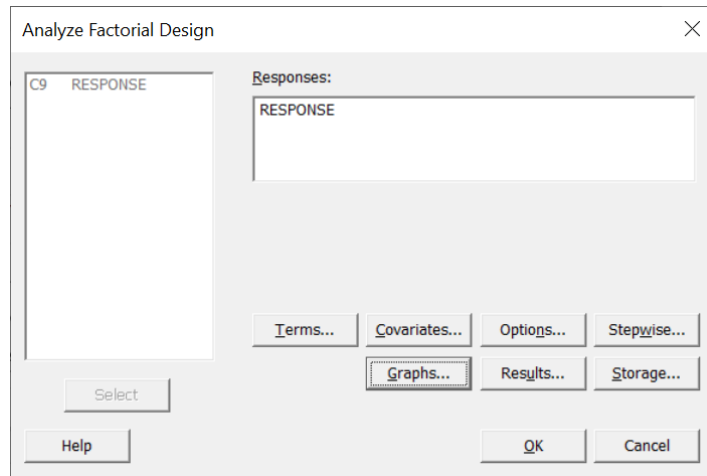
Analyze for effects

Analyze the residuals

| RUN | STD | BLOCK | CP | A | B  | C  | D  | RESPONSE |
|-----|-----|-------|----|---|----|----|----|----------|
| 1   | 1   | 1     | 1  | 1 | -1 | -1 | -1 | 6.899    |
| 2   | 2   | 2     | 1  | 1 | 1  | 1  | -1 | 8.038    |
| 3   | 3   | 3     | 1  | 1 | 1  | -1 | 1  | 13.824   |
| 4   | 4   | 4     | 1  | 1 | -1 | 1  | 1  | 11.540   |
| 5   | 5   | 5     | 1  | 1 | 1  | -1 | -1 | 9.845    |
| 6   | 6   | 6     | 1  | 1 | -1 | 1  | -1 | 7.008    |
| 7   | 7   | 7     | 1  | 1 | -1 | -1 | 1  | 10.296   |
| 8   | 8   | 8     | 1  | 1 | 1  | 1  | 1  | 9.923    |

# EXAMPLE

In Minitab



# EXAMPLE

In Minitab,

Now with fewer data points, not as many terms can be found significant that were before. Their contribution is smaller and so not visible with a fractional design.

Before all main effects were significant and the AB interaction.

Now we cannot see B or D.

## Coded Coefficients

| Term     | Effect | Coef   | SE Coef | T-Value | P-Value | VIF  |
|----------|--------|--------|---------|---------|---------|------|
| Constant |        | 9.672  | 0.201   | 48.10   | 0.000   |      |
| A        | 1.472  | 0.736  | 0.201   | 3.66    | 0.067   | 1.00 |
| B        | -1.089 | -0.544 | 0.201   | -2.71   | 0.114   | 1.00 |
| C        | 3.448  | 1.724  | 0.201   | 8.58    | 0.013   | 1.00 |
| D        | -0.807 | -0.404 | 0.201   | -2.01   | 0.182   | 1.00 |
| A*B      | -1.765 | -0.883 | 0.201   | -4.39   | 0.048   | 1.00 |

## Model Summary

| S        | R-sq   | R-sq(adj) | R-sq(pred) |
|----------|--------|-----------|------------|
| 0.568659 | 98.33% | 94.15%    | 73.24%     |

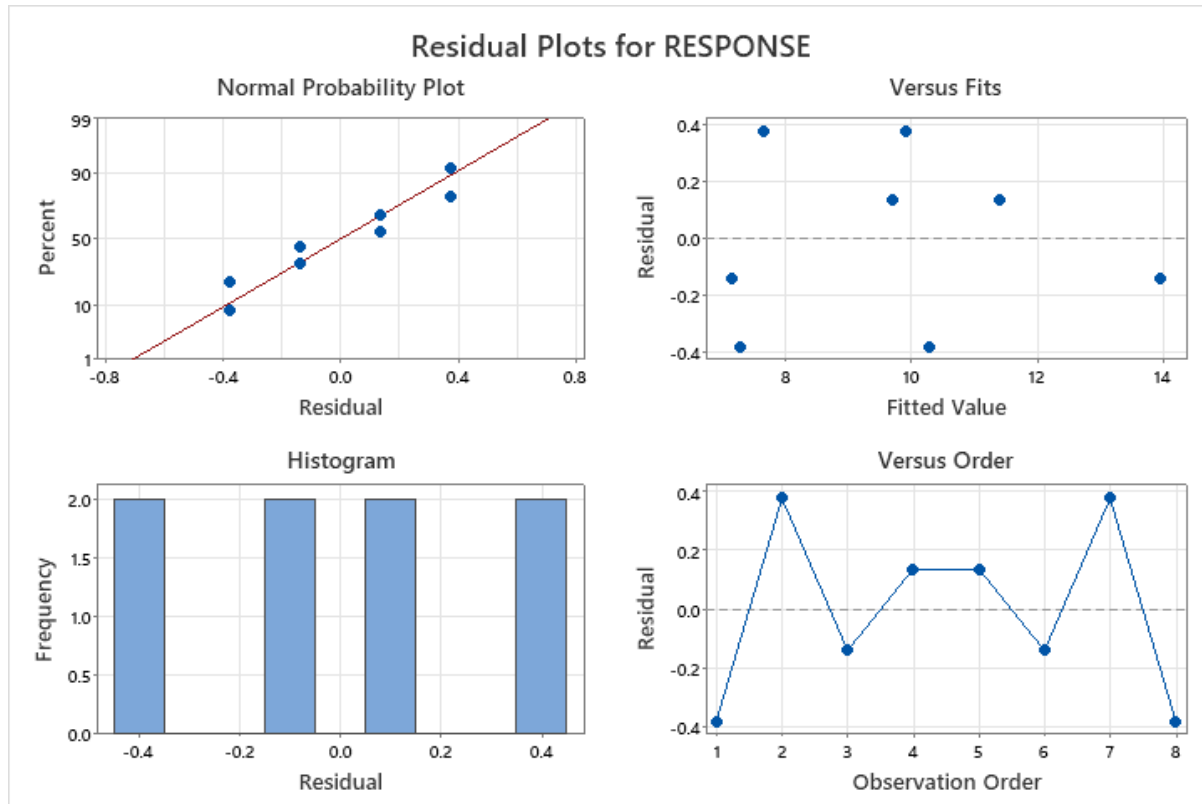
# EXAMPLE

In Minitab:

The residuals do not look random  
on the normal probability plot.

The histogram looks not normal.

On the observation order, the  
residuals are often the same  
value! OR 2X a value!



# EXAMPLE

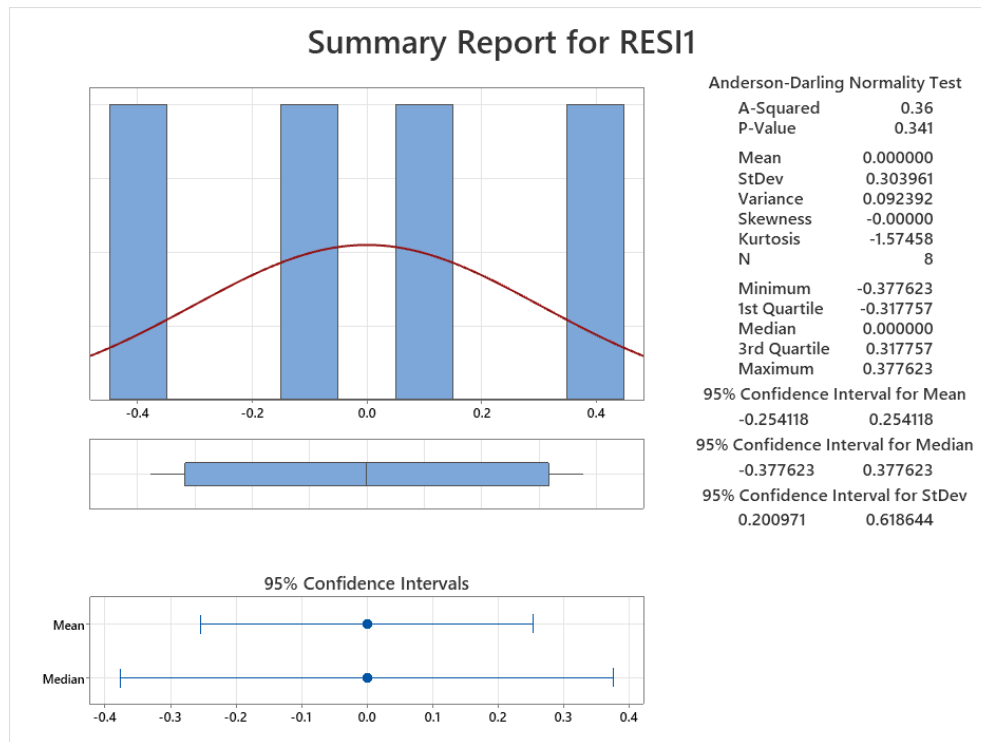
In Minitab:

AD test says residuals are normal.

Even though you see a pattern,  
Andersen Darling doesn't.

The pattern is because there are  
so few data points, the regression  
fit a plan mid-way through the  
data, and so the residuals are  
equal.

With fractional factorial designs,  
the residuals are this way. On the  
residual vs fitted value says much  
of anything.





# EXAMPLE 2.1

In Minitab:

Stat > DOE > Factorial > Define Custom Factorial Design

Stat > DOE > Factorial > Analyze Factorial Design

## Coded Coefficients

| Term     | Effect | Coef   | SE Coef | T-Value | P-Value | VIF  |
|----------|--------|--------|---------|---------|---------|------|
| Constant |        | 9.783  | 0.130   | 75.37   | 0.000   |      |
| A        | 1.214  | 0.607  | 0.130   | 4.68    | 0.001   | 1.00 |
| B        | -0.906 | -0.453 | 0.130   | -3.49   | 0.006   | 1.00 |
| C        | 3.359  | 1.680  | 0.130   | 12.94   | 0.000   | 1.00 |
| D        | -0.692 | -0.346 | 0.130   | -2.66   | 0.024   | 1.00 |
| A*B      | -1.186 | -0.593 | 0.130   | -4.57   | 0.001   | 1.00 |

## Model Summary

| S        | R-sq   | R-sq(adj) | R-sq(pred) |
|----------|--------|-----------|------------|
| 0.519159 | 95.83% | 93.74%    | 89.31%     |

C1 RUN

C2 STD

C3 BLOCK

C4 CP

C5 A

C6 B

C7 C

C8 D

C9 RESPONSE

Factors:

A-D

☒ 2-level factorial
 ☐ 2-level split-plot (hard-to-change factors)
 ☐ General full factorial

Factors from above that are hard to

Select

Low/High...

Designs...

Help

OK

Cancel

C1 RUN

C2 STD

C3 BLOCK

C4 CP

C9 RESPONSE

Standard Order Column

☐ Order of the data
 ☒ Specify by column: STD

Run Order Column

☐ Order of the data
 ☒ Specify by column: RUN

Center Points

☐ No Center Points
 ☒ Specify by column: CP

Blocks

☐ No blocks
 ☒ Specify by column: BLOCK

Select

Help

OK

Cancel

# EXAMPLE 2.1

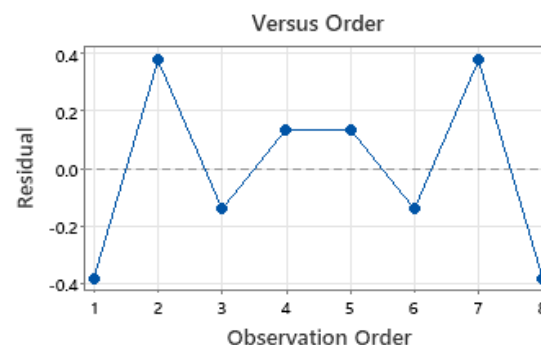
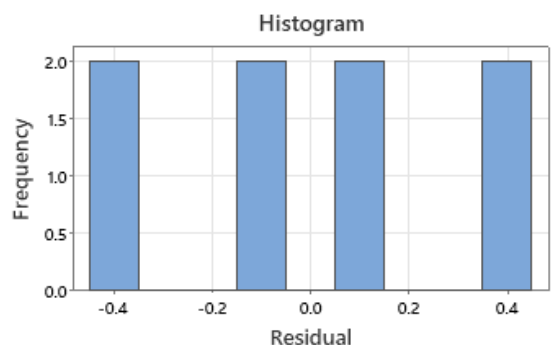
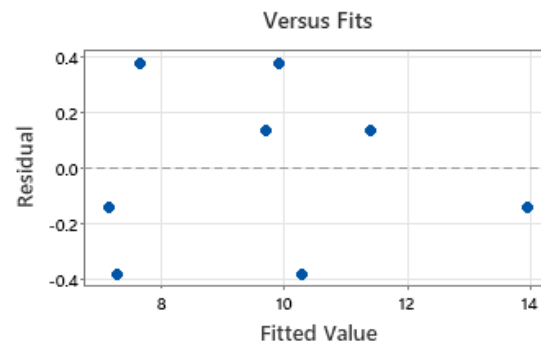
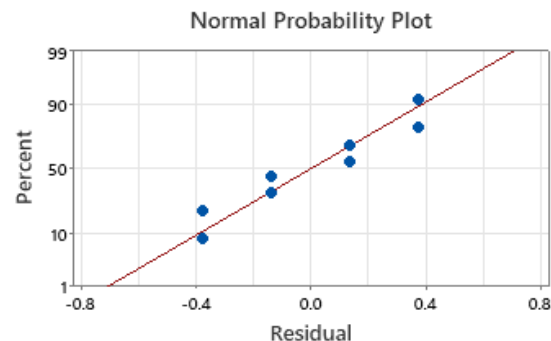
## Coded Coefficients

| Term     | Effect | Coef   | SE Coef | T-Val |
|----------|--------|--------|---------|-------|
| Constant |        | 9.672  | 0.201   | 48    |
| A        | 1.472  | 0.736  | 0.201   | 3     |
| B        | -1.089 | -0.544 | 0.201   | -2    |
| C        | 3.448  | 1.724  | 0.201   | 8     |
| D        | -0.807 | -0.404 | 0.201   | -2    |
| A*B      | -1.765 | -0.883 | 0.201   | -4    |

## Model Summary

| S        | R-sq   | R-sq(adj) | R-sq(pred) |
|----------|--------|-----------|------------|
| 0.568659 | 98.33% | 94.15%    | 73.24%     |

## Residual Plots for RESPONSE



## EXAMPLE 2.1

```
ar_list = ["x1", "x2", "x3", "x4"]
gen = 'a b c abc'
ffd = Design.from_fracfact(var_list, gen)
df_ffd = ffd.to_df()
df_ffd['Response'] = [6.899, 9.845, 7.008, 8.038,
                      10.296, 13.824, 11.540, 9.923]

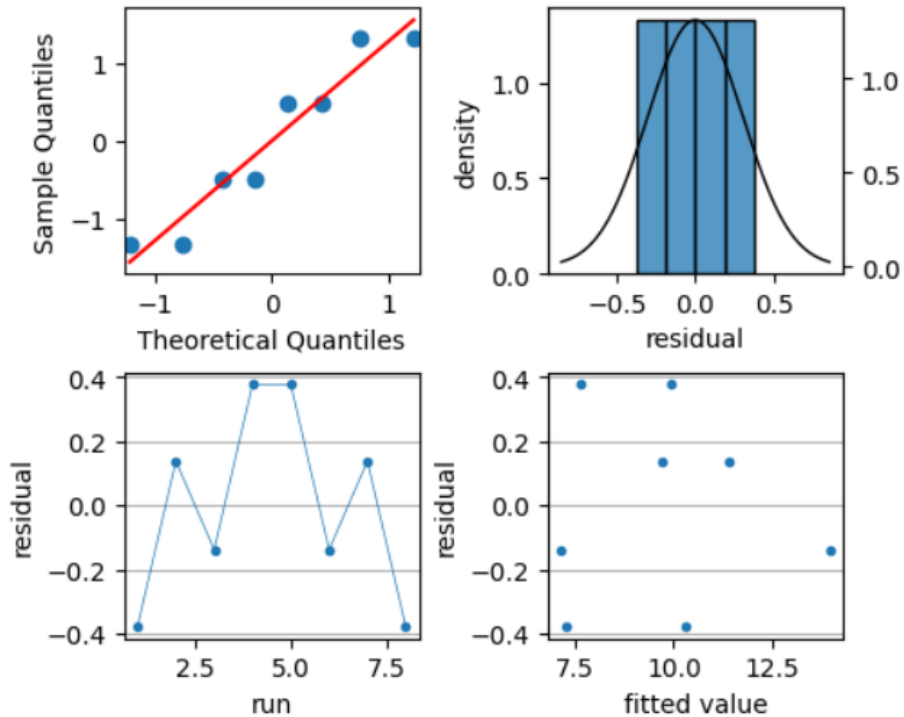
#form = 'Response ~ x1 + x2 + x3 + x4 + x1*x2 + x1*x3 + x1*x4 + x2*x3 + x2*x4 + x3*x4'
form = 'Response ~ x1 + x2 + x3 + x4 + x1*x2'
mod = smf.ols(form, df_ffd)
res = mod.fit()

with Figure(5, 4, 2, 2) as (fig, ax):
    mqr.plot.regression.residuals(res, axs=ax)
    plot = grab_figure(fig)

hstack(
    vstack(
        mqr.anova.adequacy(res),
        mqr.anova.coeffs(res)
    ),
    plot
)
```

# EXAMPLE 2.1

|           | S         | R-sq      | R-sq (adj) | N        | JB       | p (JB)   |
|-----------|-----------|-----------|------------|----------|----------|----------|
|           | 0.569205  | 0.983242  | 0.941347   | 8        | 0.668593 | 0.715842 |
|           | Coeff     | [2.5%     | 97.5%]     | PR(>F)   | VIF      |          |
| Intercept | 9.671625  | 8.805741  | 10.537509  | 0.000433 | 1.0      |          |
| x1        | 0.735875  | -0.130009 | 1.601759   | 0.067324 | 1.0      |          |
| x2        | -0.544375 | -1.410259 | 0.321509   | 0.113803 | 1.0      |          |
| x3        | 1.724125  | 0.858241  | 2.590009   | 0.013352 | 1.0      |          |
| x4        | -0.403625 | -1.269509 | 0.462259   | 0.182737 | 1.0      |          |
| x1:x2     | -0.882625 | -1.748509 | -0.016741  | 0.048255 | 1.0      |          |





# Overview

Experimentation

Effects and Interactions

Full Factorial Experiment

Fractional Factorial Designs

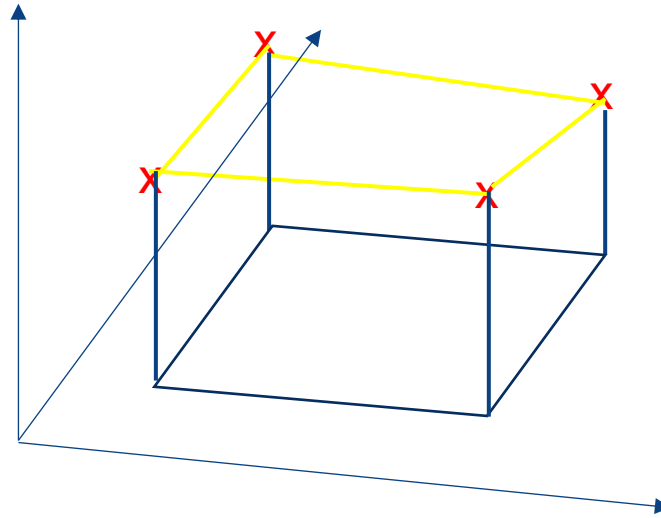
Center Points

Central Composite Designs

Computer Experiments and Latin Hypercubes

# Center Points

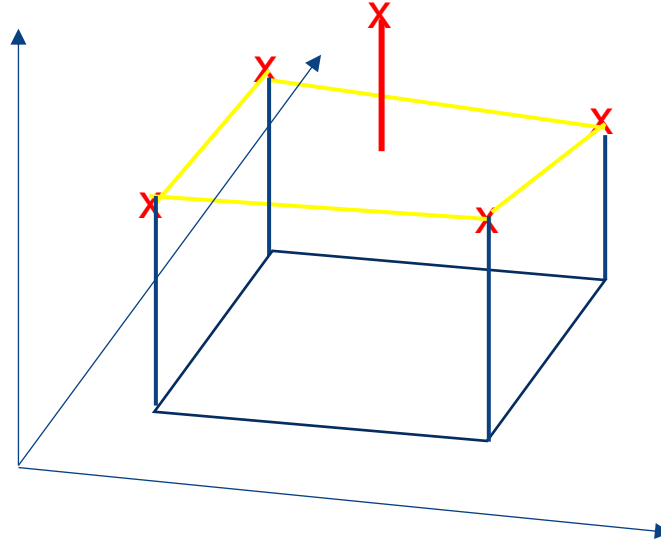
Factorial experiment measure the response trends at the corners only, and consider the center as the average of the measured responses.



# Center Points

Factorial experiment measure the response trends at the corners only, and consider the center as the average of the measured responses.

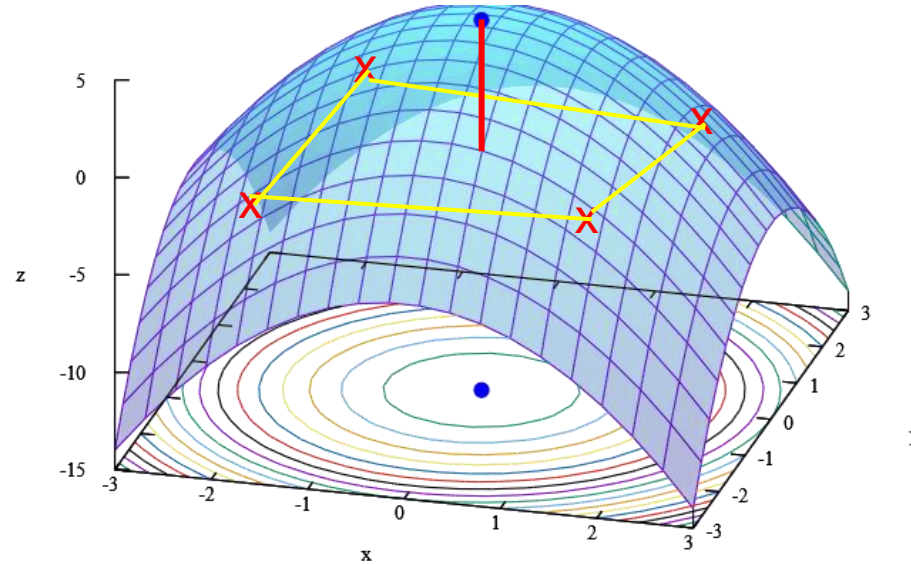
The center might be offset quadratically.



# Center Points

Factorial experiment measure the response trends at the corners only, and consider the center as the average of the measured responses.

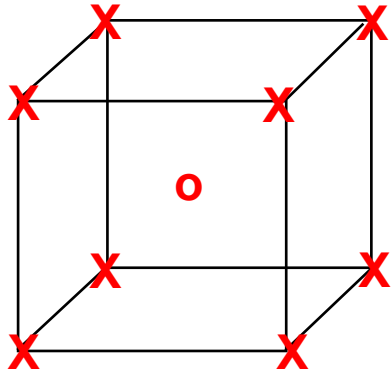
The center might be offset quadratically.





# Center Points

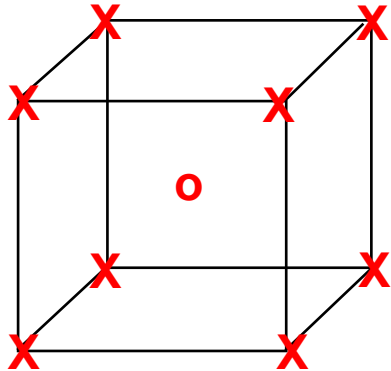
To test whether a more complex model is needed, center points can be experimented at. If there is quadratic behavior, a single center point run will show it. An efficient test.



| $X_1$ | $X_2$ | $X_3$ | $X_4$ |
|-------|-------|-------|-------|
| -1    | -1    | -1    | -1    |
| 1     | -1    | -1    | -1    |
| -1    | 1     | -1    | 1     |
| 1     | 1     | -1    | 1     |
| -1    | -1    | 1     | 1     |
| 1     | -1    | 1     | 1     |
| -1    | 1     | 1     | -1    |
| 1     | 1     | 1     | -1    |
| 0     | 0     | 0     | 0     |

# Center Points

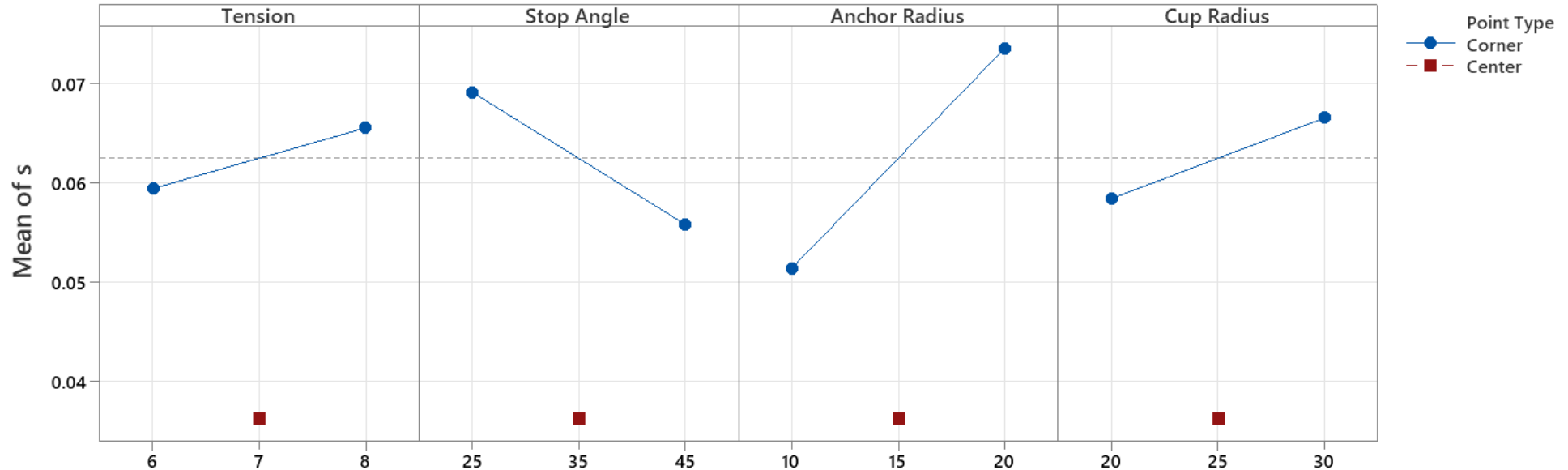
To test whether a more complex model is needed, center points can be experimented at. Doing replicates of the center point can provide an estimate of the experimental error more efficiently than replicates at the corner points.



| $X_1$ | $X_2$ | $X_3$ | $X_4$ |
|-------|-------|-------|-------|
| -1    | -1    | -1    | -1    |
| 1     | -1    | -1    | -1    |
| -1    | 1     | -1    | 1     |
| 1     | 1     | -1    | 1     |
| -1    | -1    | 1     | 1     |
| 1     | -1    | 1     | 1     |
| -1    | 1     | 1     | -1    |
| 1     | 1     | 1     | -1    |
| 0     | 0     | 0     | 0     |
| 0     | 0     | 0     | 0     |
| 0     | 0     | 0     | 0     |

# Center Points

To test whether a more complex model is needed, center points can be experimented at. If there is quadratic behaviour, a center point offset will show.



## Example

Suppose you have 4 process variables whose mean shifts you suspect contribute to the mean shift of the process output response. Of the four (A,B,C,D), you suspect there is an interact among BC.

Design a two factorial experiment with a minimal number of runs.

# Example

Four process variables (A,B,C,D) are candidates as significant sources of means shifts for two different response KPIs. Analyze the following experimental results on two different responses. What is the best fit response equation for each?

| A  | B  | C  | D  | Y1    | Y2    |
|----|----|----|----|-------|-------|
| -1 | -1 | -1 | -1 | 95.3  | 98.1  |
| 1  | -1 | -1 | -1 | 105.0 | 108.2 |
| -1 | 1  | -1 | 1  | 89.0  | 92.5  |
| 1  | 1  | -1 | 1  | 99.4  | 102.3 |
| -1 | -1 | 1  | 1  | 105.1 | 108.5 |
| 1  | -1 | 1  | 1  | 115.2 | 118.3 |
| -1 | 1  | 1  | -1 | 91.0  | 94.1  |
| 1  | 1  | 1  | -1 | 101.1 | 104.1 |
| 0  | 0  | 0  | 0  | 100.4 | 100.2 |

## EXERCISE: 2.2: Y<sub>1</sub>

| CenterPt | A  | B  | C  | D  | Y1 | Y2      |
|----------|----|----|----|----|----|---------|
| 1        | -1 | -1 | -1 | -1 | -1 | 100.274 |
| 1        | 1  | -1 | -1 | -1 | 1  | 104.418 |
| 1        | -1 | 1  | -1 | -1 | 1  | 84.956  |
| 1        | 1  | 1  | -1 | -1 | -1 | 99.206  |
| 1        | -1 | -1 | 1  | 1  | 1  | 100.162 |
| 1        | 1  | -1 | 1  | 1  | -1 | 115.706 |
| 1        | -1 | 1  | 1  | 1  | -1 | 96.436  |
| 1        | 1  | 1  | 1  | 1  | 1  | 99.8    |
| 0        | 0  | 0  | 0  | 0  | 0  | 99.527  |

### Coded Coefficients

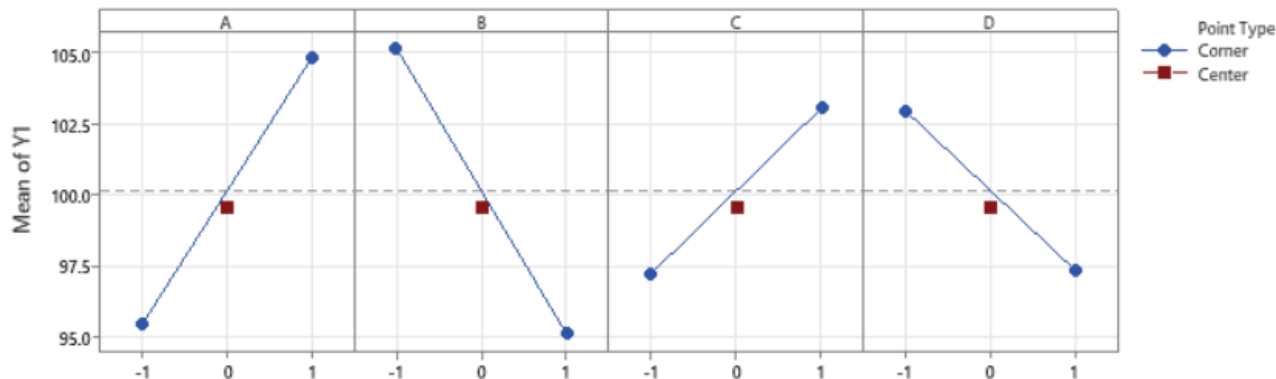
| Term     | Effect  | Coef    | SE Coef | T-Value | P-Value | VIF  |
|----------|---------|---------|---------|---------|---------|------|
| Constant |         | 100.120 | 0.167   | 598.55  | 0.000   |      |
| A        | 9.325   | 4.663   | 0.167   | 27.88   | 0.000   | 1.00 |
| B        | -10.040 | -5.020  | 0.167   | -30.01  | 0.000   | 1.00 |
| C        | 5.813   | 2.906   | 0.167   | 17.37   | 0.000   | 1.00 |
| D        | -5.572  | -2.786  | 0.167   | -16.65  | 0.000   | 1.00 |
| Ct Pt    |         | -0.593  | 0.502   | -1.18   | 0.323   | 1.00 |

### Model Summary

| S        | R-sq   | R-sq(adj) | R-sq(pred) |
|----------|--------|-----------|------------|
| 0.473114 | 99.87% | 99.65%    | *          |

### Main Effects Plot for Y<sub>1</sub>

Fitted Means



# EXERCISE: 2.2: Y1

```
var_list = ["A", "B", "C", "D"]
gen = 'a b c abc'
nc = 1 # number of centerpoints

cfd = Design.from_fracfact(var_list, gen) + Design.from_centrepoints(var_list, nc)
df_cfd = cfd.to_df()
```

```
df_cfd['Y1'] = [100.274, 104.418, 84.956, 99.206, 100.162, 115.706, 96.436, 99.8, 99.527]
df_cfd['Y2'] = [100.274, 104.418, 84.956, 99.206, 100.162, 115.706, 96.436, 99.8, 97.527]
```

```
Ylabel = 'Y1'
```

```
cornerdata = df_cfd.query('PtType == 1')
centerdata = df_cfd.query('PtType == 0')
Ncorner = cornerdata.shape[0]
Ncenter = centerdata.shape[0]
CornerAvg = np.average(cornerdata[Ylabel])
CenterAvg = np.average(centerdata[Ylabel])
delta = CornerAvg - CenterAvg
```

```
form = 'Y1 ~ A + B + C + D'
mod = smf.ols(form, cornerdata)
res = mod.fit()
```

```
fitY1 = mgr.anova.adequacy(res)
ssY1 = mgr.anova.summary(res)
```

```
from scipy.stats import f
```

```
SScenter = np.power(delta,2)
dfcenter = Ncenter
SSerror = ssY1['sum_sq']['Residual']
dferror = ssY1['df']['Residual'] + Ncenter-1
if(Ncenter > 1) :
    SSerror = SSerror + np.var(centerdata['Y1'])*(Ncenter-1)
```

```
MScenter = SScenter/(dfcenter)
MSerror = SSerror/(dferror)
Fcenter = MScenter/MSerror
Pcenter = 1.0 - f.cdf(Fcenter, 1, dferror)
```

```
print('Curvature F-stat:',Fcenter,'and p =',Pcenter)
```

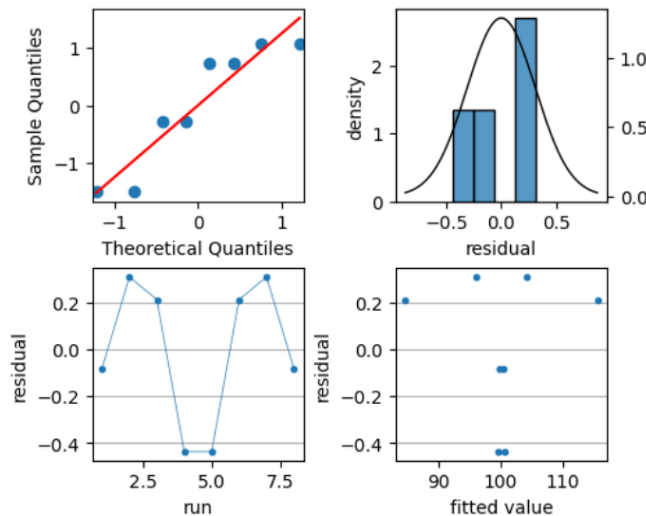
```
Curvature F-stat: 1.569683954582964 and p = 0.29901865655598525
```

# EXERCISE: 2.2: Y1

```
with Figure(5, 4, 2, 2) as (fig, ax):
    mqr.plot.regression.residuals(res, ax)
    plot = grab_figure(fig)

vstack(
    hstack(
        mqr.anova.summary(res),
        mqr.anova.coeffs(res)
    ),
    hstack(
        plot,
        vstack(
            mqr.anova.adequacy(res),
            mqr.inference.dist.test_1sample(res.resid)
        )
    )
)
```

|                 | df  | sum_sq     | mean_sq    | F          | PR(>F)   |                  | Coeff     | [2.5%     | 97.5%]     | PR(>F)       | VIF |
|-----------------|-----|------------|------------|------------|----------|------------------|-----------|-----------|------------|--------------|-----|
| <b>A</b>        | 1.0 | 173.929901 | 173.929901 | 777.039940 | 0.000101 | <b>Intercept</b> | 100.11975 | 99.587419 | 100.652081 | 1.028415e-08 | 1.0 |
| <b>B</b>        | 1.0 | 201.623281 | 201.623281 | 900.761406 | 0.000081 | <b>A</b>         | 4.66275   | 4.130419  | 5.195081   | 1.013438e-04 | 1.0 |
| <b>C</b>        | 1.0 | 67.570313  | 67.570313  | 301.873522 | 0.000416 | <b>B</b>         | -5.02025  | -5.552581 | -4.487919  | 8.124995e-05 | 1.0 |
| <b>D</b>        | 1.0 | 62.083225  | 62.083225  | 277.359700 | 0.000471 | <b>C</b>         | 2.90625   | 2.373919  | 3.438581   | 4.155068e-04 | 1.0 |
| <b>Residual</b> | 3.0 | 0.671510   | 0.223837   | NaN        | NaN      | <b>D</b>         | -2.78575  | -3.318081 | -2.253419  | 4.712999e-04 | 1.0 |
| <b>Total</b>    | 7.0 | 505.878228 | 505.430555 | NaN        | NaN      |                  |           |           |            |              |     |



| S        | R-sq     | R-sq (adj) | N | JB       | p (JB)   |
|----------|----------|------------|---|----------|----------|
| 0.473114 | 0.998673 | 0.996903   | 8 | 0.875299 | 0.645552 |

## Hypothesis Test

non-normality

method Anderson-Darling

**H<sub>0</sub>** distribution == normal

**H<sub>1</sub>** distribution != normal

statistic 0.537152

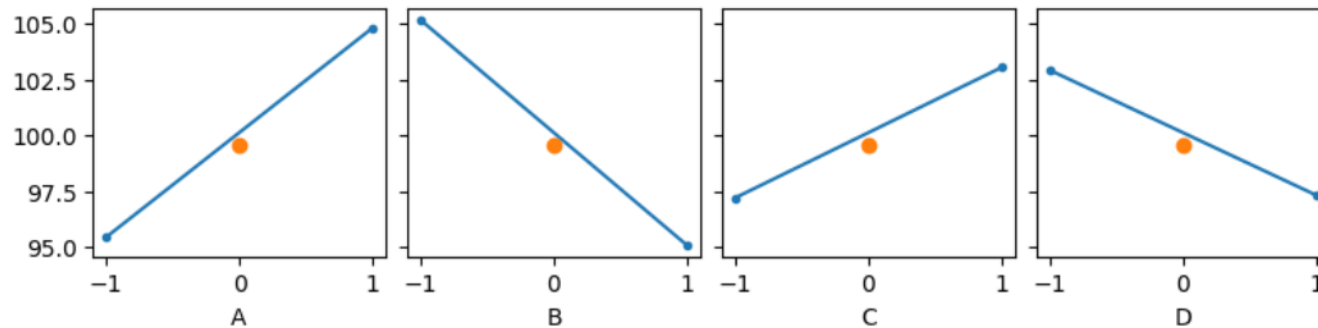
p-value 0.115168



## EXERCISE: 2.2: Y1

```
effect = Ylabel

with Figure(8, 2, 1, 4, sharey=True) as (fig, axs):
    for i, name in enumerate(cfd.names):
        axs[i].plot(cornerdata.groupby(name).mean()[effect], color='C0', marker='.')
        axs[i].plot(centerdata.groupby(name).mean()[effect], color='C1', marker='o')
        axs[i].set_xlabel(name)
```



# EXERCISE 2.2: Y2

```
from scipy.stats import f

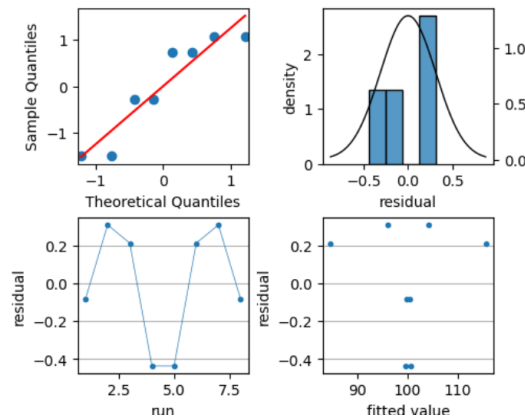
SScenter = np.power(delta,2)
dfcenter = Ncenter
SSerror = ssY1['sum_sq']['Residual']
dferror = ssY1['df']['Residual'] + Ncenter-1
if(Ncenter > 1) :
    SSerror = SSerror + np.var(centerdata['Y1'])*(Ncenter-1)

MScenter = SScenter/(dfcenter)
MSerror = SSerror/(dferror)
Fcenter = MScenter/MSerror
Pcenter = 1.0 - f.cdf(Fcenter, 1, dferror)

print('Curvature H0: Curvature == 0, no curvature.')
print('Curvature Ha: Curvature != 0, there is curvature.')
print('Curvature F-stat:',Fcenter,'and p =',Pcenter)
```

Curvature H0: Curvature == 0, no curvature.  
 Curvature Ha: Curvature != 0, there is curvature.  
 Curvature F-stat: 30.032423498848576 and p = 0.011948952989685346

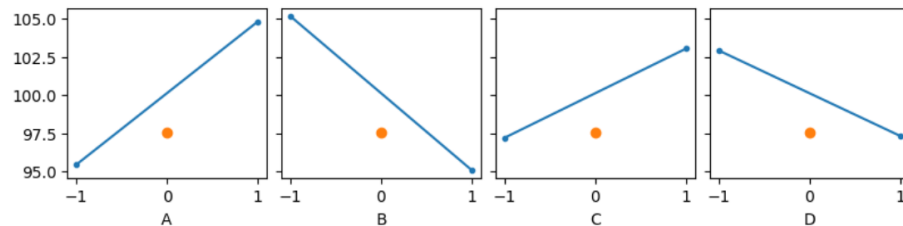
|          | df  | sum_sq     | mean_sq    | F          | PR(>F)   |           | Coeff     | [2.5%     | 97.5%]     | PR(>F)       | VIF |
|----------|-----|------------|------------|------------|----------|-----------|-----------|-----------|------------|--------------|-----|
| A        | 1.0 | 173.929901 | 173.929901 | 777.039940 | 0.000101 | Intercept | 100.11975 | 99.587419 | 100.652081 | 1.028415e-08 | 1.0 |
| B        | 1.0 | 201.623281 | 201.623281 | 900.761406 | 0.000081 | A         | 4.66275   | 4.130419  | 5.195081   | 1.013438e-04 | 1.0 |
| C        | 1.0 | 67.570313  | 67.570313  | 301.873522 | 0.000416 | B         | -5.02025  | -5.552581 | -4.487919  | 8.124995e-05 | 1.0 |
| D        | 1.0 | 62.083225  | 62.083225  | 277.359700 | 0.000471 | C         | 2.90625   | 2.373919  | 3.438581   | 4.155068e-04 | 1.0 |
| Residual | 3.0 | 0.671510   | 0.223837   | NaN        | NaN      | D         | -2.78575  | -3.318081 | -2.253419  | 4.712999e-04 | 1.0 |
| Total    | 7.0 | 505.878228 | 505.430555 | NaN        | NaN      |           |           |           |            |              |     |



| S        | R-sq     | R-sq (adj) | N | JB       | p (JB)   |
|----------|----------|------------|---|----------|----------|
| 0.473114 | 0.998673 | 0.996903   | 8 | 0.875299 | 0.645552 |

Hypothesis Test  
non-normality

|                |                        |
|----------------|------------------------|
| method         | Anderson-Darling       |
| H <sub>0</sub> | distribution == normal |
| H <sub>1</sub> | distribution != normal |
| statistic      | 0.537152               |
| p-value        | 0.115168               |





# Overview

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Full Factorial Experiment

Fractional Factorial Designs

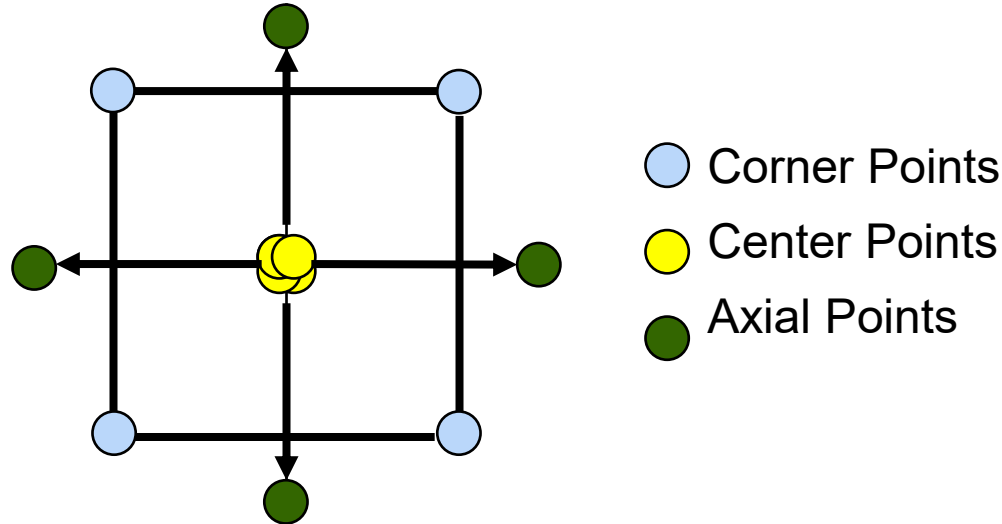
Center Points

Central Composite Designs

Computer Experiments and Latin Hypercubes

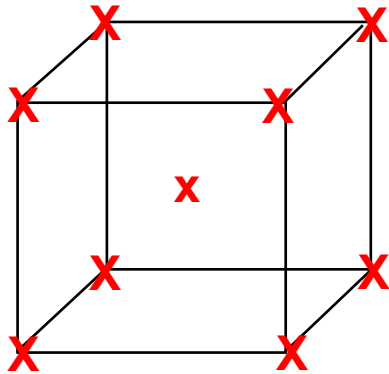
# Response Surface Designs

If center points indicate quadratic terms, then axial points can be added to the two factorial analysis. Axial points allow  $x^2$  quadratic terms to be solved for.

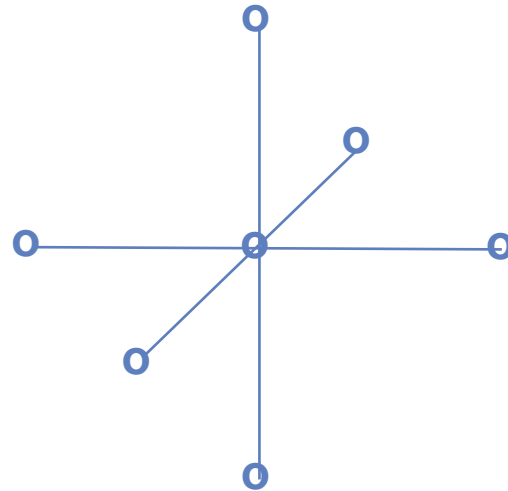


# Central Composite Designs

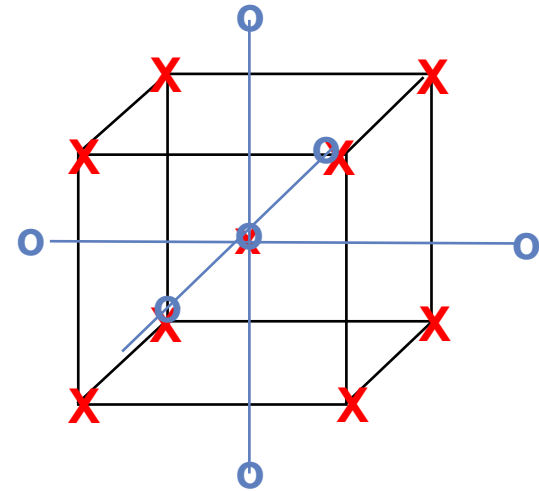
A **central composite design** is an augmentation of a two factorial design to allow quadratic and interaction terms in the response equation. More efficient than 3+ factorial designs.



+



=



Factorial Design

+

Axial Points

=

CCD Design

# CCD in Minitab

In Minitab, **stat > DOE > Response Surface Design**

Create Response Surface Design

Type of Design

☒ Central composite (2 to 10 continuous factors)

☐ Box-Behnken (3,4,5,6,7,9, or 10 continuous factors)

Number of continuous factors: 4

Number of categorical factors: 0

Display Available Designs...

Designs... Factors...

Options... Results...

Help OK Cancel

Create Response Surface Design: Designs

| Designs | Runs | Blocks | Center Points |      |       | Default Alpha |
|---------|------|--------|---------------|------|-------|---------------|
|         |      |        | Total         | Cube | Axial |               |
| Full    | 31   | 1      | 7             | 0    | 0     | 2.000         |
| Full    | 30   | 2      | 6             | 4    | 2     | 2.000         |
| Full    | 30   | 3      | 6             | 4    | 2     | 2.000         |

Number of Center Points

☒ Default

☐ Custom

Cube block: Axial block:

Value of Alpha

☒ Default

☐ Face Centered

☐ Custom:

Number of replicates: 1

☐ Block on replicates

Help OK Cancel

# CCD in Python

Central composite designs can be created in Python

```
X = ccdesign(n, center, alpha, face)
```

- Where `n` is the number of variables, `center` indicates the number of center points, and `alpha` and `face` specify the CCD type.
- The result is in 'coded units' (-1, 1)
- Create a second 'uncoded units' matrix replacing these with the levels of each variable
- The factorial portion is only available in full factorial. Custom construction is needed for fractional factorial CCD designs.

```
>> from pyDOE2 import *  
>> X = ccdesign(3, center=(3,3),  
               alpha=al, face=ff)
```

```
    X =  
[ -1.      , -1.      , -1.      ],  
[  1.      , -1.      , -1.      ],  
[ -1.      ,  1.      , -1.      ],  
[  1.      ,  1.      , -1.      ],  
[ -1.      , -1.      ,  1.      ],  
[  1.      , -1.      ,  1.      ],  
[ -1.      ,  1.      ,  1.      ],  
[  1.      ,  1.      ,  1.      ],  
[  0.      ,  0.      ,  0.      ],  
[  0.      ,  0.      ,  0.      ],  
[  0.      ,  0.      ,  0.      ],  
[ -1.80906807,  0.      ,  0.      ],  
[  1.80906807,  0.      ,  0.      ],  
[  0.      , -1.80906807,  0.      ],  
[  0.      ,  1.80906807,  0.      ],  
[  0.      ,  0.      , -1.80906807],  
[  0.      ,  0.      ,  1.80906807],  
[  0.      ,  0.      ,  0.      ],  
[  0.      ,  0.      ,  0.      ],  
[  0.      ,  0.      ,  0.      ]
```

# Available RSM Designs

The central composite designs recommended are shown by Minitab

Create Response Surface Design: Display Available Designs

Available Response Surface Designs

| Design                    |           | Continuous Factors |    |    |    |    |     |     |     |     |
|---------------------------|-----------|--------------------|----|----|----|----|-----|-----|-----|-----|
|                           |           | 2                  | 3  | 4  | 5  | 6  | 7   | 8   | 9   | 10  |
| Central composite full    | unblocked | 13                 | 20 | 31 | 52 | 90 | 152 |     |     |     |
|                           | blocked   | 14                 | 20 | 30 | 54 | 90 | 160 |     |     |     |
| Central composite half    | unblocked |                    |    |    | 32 | 53 | 88  | 154 |     |     |
|                           | blocked   |                    |    |    | 33 | 54 | 90  | 160 |     |     |
| Central composite quarter | unblocked |                    |    |    |    |    |     | 90  | 156 |     |
|                           | blocked   |                    |    |    |    |    |     | 90  | 160 |     |
| Central composite eighth  | unblocked |                    |    |    |    |    |     |     |     | 158 |
|                           | blocked   |                    |    |    |    |    |     |     |     | 160 |
| Box-Behnken               | unblocked |                    | 15 | 27 | 46 | 54 | 62  |     | 130 | 170 |
|                           | blocked   |                    |    | 27 | 46 | 54 | 62  |     | 130 | 170 |

Help

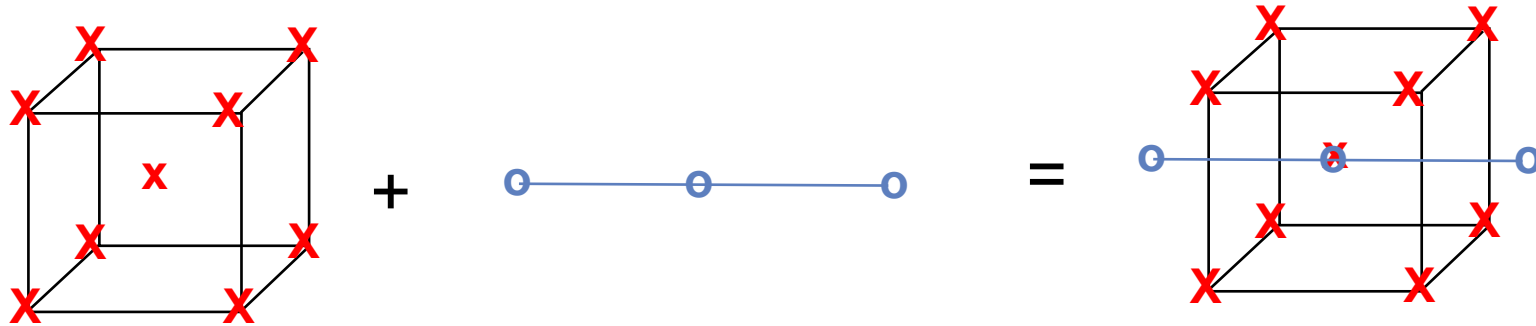
OK



# Custom RSM Designs

If you know which variable is causing the quadratic behaviour, you can expand the axial points on that variable alone, rather than all variables.

This reduces the needed experimentation.



Statistics packages won't generate them and perhaps won't even easily analyze them.

This starts getting into the art of design-of-experiments.



# Other Experimental Designs

There are many experimental designs

- Plackett-Burman designs
- 2 level factorial designs
- 3 level factorial designs
- N level factorial designs
- Mixture designs
- Taguchi designs
- Box-Behnken designs
- Central Composite designs
- D-optimal designs

These all need more detailed discussions.

In general, 2-factorial to CCD designs are sufficient for many manufacturing applications.



# Factorial Design Summary

Full factorial analysis is inefficient in most cases

Two level factorial designs represent main effects and interactions

- Not quadratic terms

Fractional factorial designs confound terms

Regression is used to determine significant terms and establish a simplified equation

To check for quadratic non-linearity, a center point can be added

If there is significant quadratic non-linearity, axial points can be added as a central composite design. This sequential expansion is most efficient.



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Effects and Interactions

Full Factorial Experiment

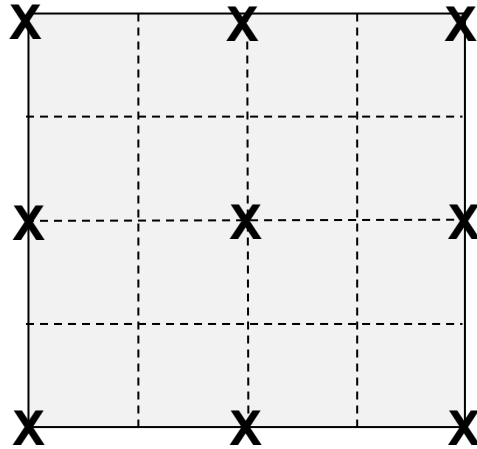
Yates Labeling

Fractional Factorial Designs

Computer Experiments and Latin Hypercubes

# Design of Experiments

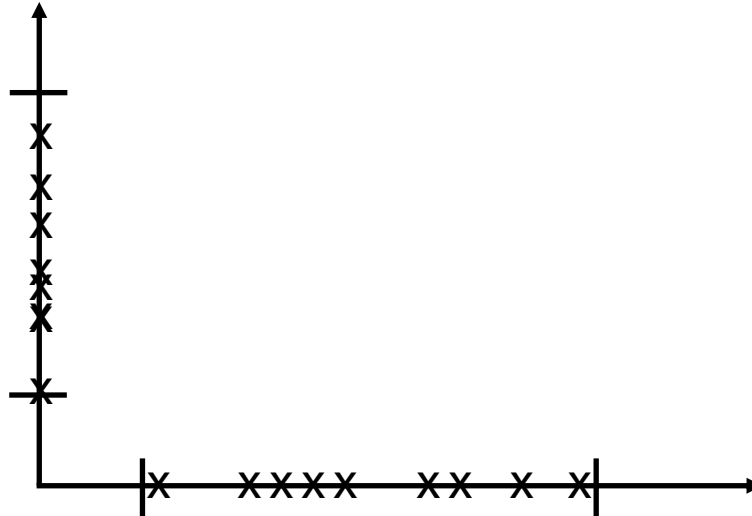
With factorial design, we pick sample points in a factorial manner.





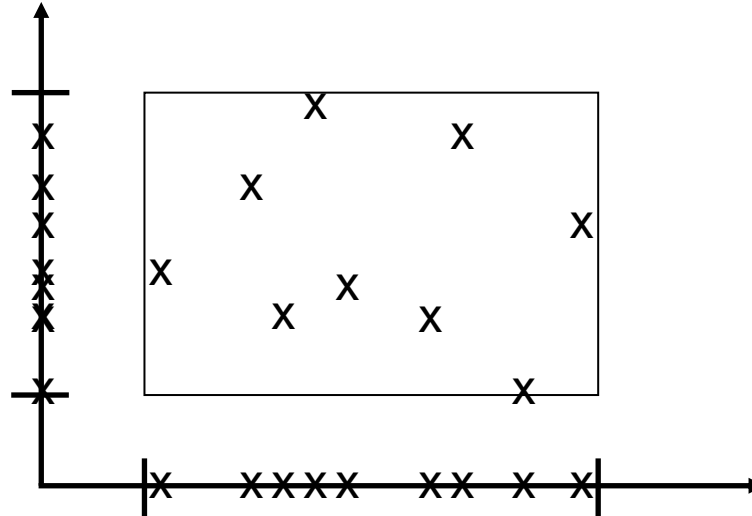
# Monte-Carlo Simulation

A different approach is to consider each variable and randomly sample points within the interval



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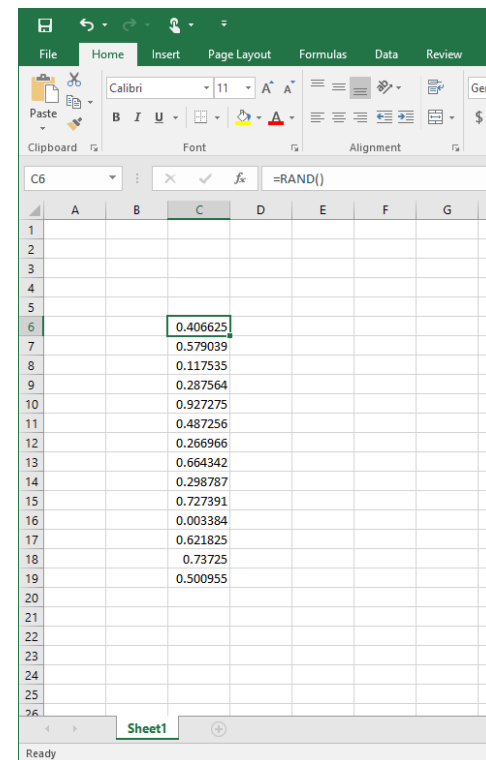


This random sampling approach is also called **Monte-Carlo** sampling. Notice there are random locations of high and low density of points. We can probably do better.

# Random Number generation

Computers easily generate uniform distribution samples in coded units between 0 and 1.

- In Excel the `rand()` function generates a single random value
- In Python `np.random.random_sample()`







## Convert to Unencoded Units

Suppose you generate random variables, each  $(p_1 \ p_2 \ \dots \ p_D) = \vec{p} \in [0,1]^D$ .

Transform each  $p_i$  from  $[0,1]$  to a real valued using the inverse distribution function.

$$x_i = \varphi^{-1}(p_i)$$

For example, if  $x_i$  is normally distributed with mean  $\mu$  and sigma  $\sigma$ ,

$$x_i = \text{norminv}(p_i, \mu, \sigma)$$

For example, if  $x_i$  is uniformly distributed within  $[A,B]$ ,

$$x_i = A + p_i(B - A)$$

# Monte-Carlo Experimentation – Excel

Create a design matrix  $X = (x_1, x_2)$  using rand() and the cumulative inverse function.  
The create a  $Y$  column evaluated at each  $X$  row, and use regression.

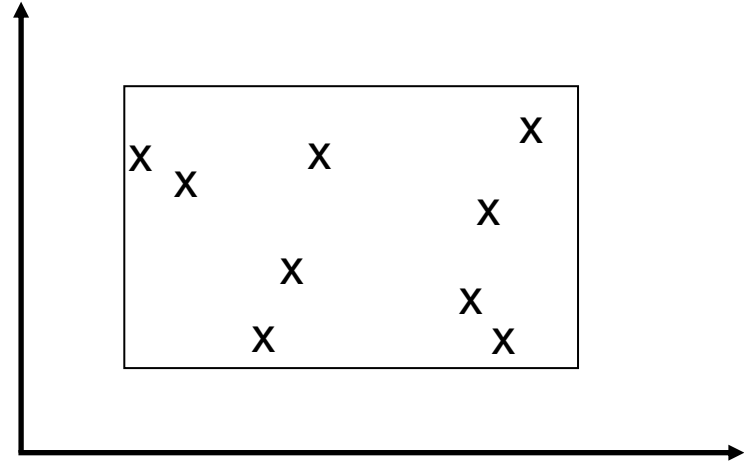
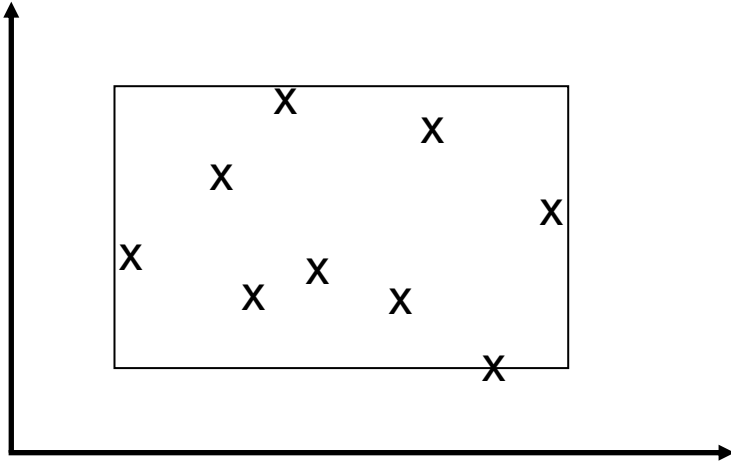
|          |          |          |          |
|----------|----------|----------|----------|
|          | mean     | 10       | 20       |
|          | sigma    | 1        | 2        |
| Cu1      | Cu2      | X1       | X2       |
| 0.689434 | 0.245081 | 10.49425 | 18.6199  |
| 0.196944 | 0.067239 | 9.147412 | 17.00665 |
| 0.748072 | 0.805779 | 10.66844 | 21.72489 |
| 0.114294 | 0.693811 | 8.795998 | 21.01336 |
| 0.448948 | 0.281124 | 9.871681 | 18.84099 |
| 0.254579 | 0.454109 | 9.339851 | 19.76942 |
| 0.644758 | 0.993016 | 10.37121 | 24.91621 |
| 0.156882 | 0.374594 | 8.992645 | 19.36058 |
| 0.39986  | 0.348017 | 9.746291 | 19.21864 |
| 0.708084 | 0.805292 | 10.54779 | 21.72135 |

Cu=rand()

=norm.inv(Cu,mu,s)

# Latin Hypercube Sampling

Random sampling is not the most efficient layout of points



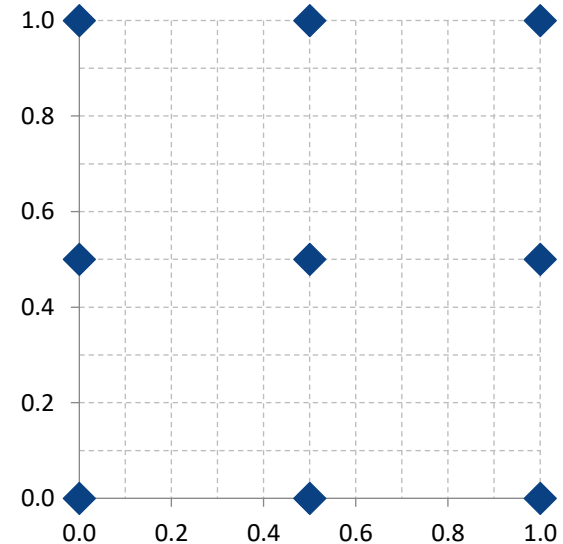
Which random sample is better?

# Latin Hypercube Sampling

Traditional DOE is also not the most efficient layout of points

There are  $m < N$  levels on each variable.

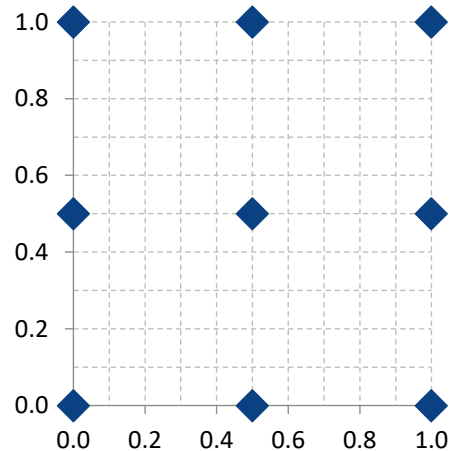
- Very good for fabricated part experiments.
- Not good for regression fitting many terms.



# Latin Hypercube Sampling

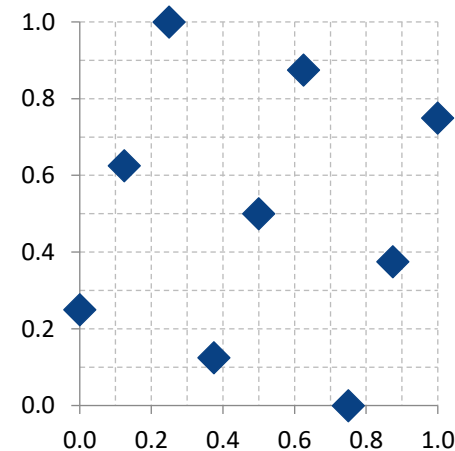
A slightly rotated set of points provides many more levels

9 Run 3 Factorial Design



3 levels for each variable  
Replicated points on each projection  
At most, quadratic effects captured

9 Run Latin Hypercube



9 levels for each variable  
No replicates on each projection  
High order effects can be captured

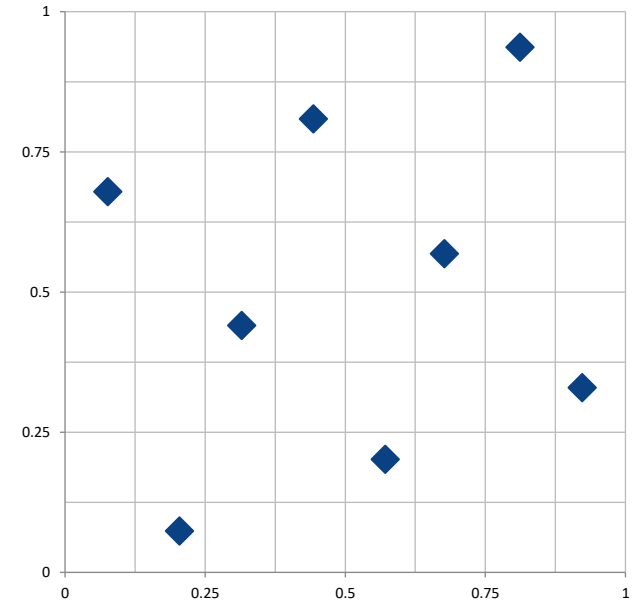
# Latin Hypercube Sampling

Operates on a  $d$ -dimensional  $[0,1]^d$  hypercube space

For each of  $d$  variables, split the interval into  $N$  bins

Pick  $N$  cells out of the  $N^d$  such that each bin is selected only once on each variable

Each point in the space define a vector, each with unique values on each variable

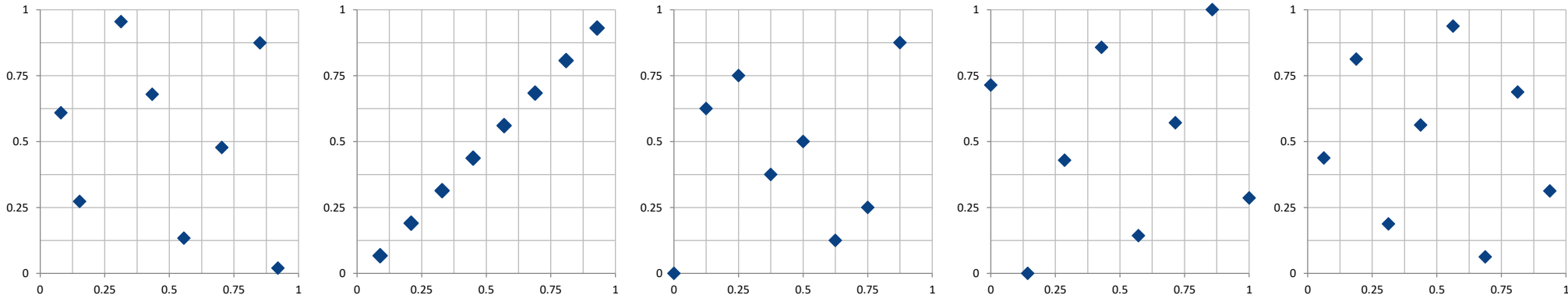


# Non-Unique

There are many ways to construct a valid Latin Hypercube with  $N$  points and  $d$  variables.

Good ones have high **discrepancy** – they fill the space evenly.

Generally, finding a good Latin hypercube is an iterative optimization.



Each of these are 2-dimensional 8-point Latin Hypercubes



# Generating Latin Hypercubes

Not available in Excel or Minitab

Python: using the library pyDOE

```
lhs(d, samples, criterion, iterations)
```

- d = number of variables
- samples = N

Use the Latin hypercube as the encoded form of the design matrix.

Then each run can be computed for a response, and regression applied.

```
>> lhs(4, samples=10, criterion='center')
array([[ 0.05,  0.05,  0.15,  0.15],
       [ 0.55,  0.85,  0.95,  0.75],
       [ 0.25,  0.25,  0.45,  0.25],
       [ 0.45,  0.35,  0.75,  0.45],
       [ 0.75,  0.55,  0.25,  0.55],
       [ 0.95,  0.45,  0.35,  0.05],
       [ 0.35,  0.95,  0.05,  0.65],
       [ 0.15,  0.65,  0.55,  0.35],
       [ 0.85,  0.75,  0.85,  0.85],
       [ 0.65,  0.15,  0.65,  0.95]])
```





# Summary

Design of experiments offers the means to cover a domain efficiently for model fitting

Full factorial experiments are inefficient

Fractional factorial designs offer less runs but care must be taken to consider confounding

Center points can indicate non-linearity. Fractional factorial + center points are the recommended first designed experiment to undertake in most manufacturing problems.

Central composite designs are an efficient way to include non-linearities

Computer experiments offer means to study more complex experimental layouts. Latin hypercubes are one basic form of computer based experiments. Other methods are more advanced with various properties such as sequence based sampling.

# Example

The previous example with process variables (A,B,C,D) was expanded into a central composite design with a second axial point experiment. Analyze the following results on two different responses. What is the best fit response equation for each?

The PtType and Blocks are useful to Minitab in defining this custom response surface experiment.

| Run | PtType | Blocks | A     | B     | C     | D     | Y1    | Y2    |
|-----|--------|--------|-------|-------|-------|-------|-------|-------|
| 1   | 1      | 1      | -1    | -1    | -1    | -1    | 95.3  | 98.1  |
| 2   | 1      | 1      | 1     | -1    | -1    | -1    | 105.0 | 108.2 |
| 3   | 1      | 1      | -1    | 1     | -1    | 1     | 89.0  | 92.5  |
| 4   | 1      | 1      | 1     | 1     | -1    | 1     | 99.4  | 102.3 |
| 5   | 1      | 1      | -1    | -1    | 1     | 1     | 105.1 | 108.5 |
| 6   | 1      | 1      | 1     | -1    | 1     | 1     | 115.2 | 118.3 |
| 7   | 1      | 1      | -1    | 1     | 1     | -1    | 91.0  | 94.1  |
| 8   | 1      | 1      | 1     | 1     | 1     | -1    | 101.1 | 104.1 |
| 9   | 0      | 1      | 0     | 0     | 0     | 0     | 100.4 | 100.2 |
| 10  | -1     | 2      | -1.41 | 0     | 0     | 0     | 93.0  | 93.1  |
| 11  | -1     | 2      | 1.41  | 0     | 0     | 0     | 107.4 | 107.2 |
| 12  | -1     | 2      | 0     | -1.41 | 0     | 0     | 105.8 | 115.9 |
| 13  | -1     | 2      | 0     | 1.41  | 0     | 0     | 94.8  | 104.6 |
| 14  | -1     | 2      | 0     | 0     | -1.41 | 0     | 95.9  | 91.9  |
| 15  | -1     | 2      | 0     | 0     | 1.41  | 0     | 104.4 | 100.6 |
| 16  | -1     | 2      | 0     | 0     | 0     | -1.41 | 97.4  | 97.2  |
| 17  | -1     | 2      | 0     | 0     | 0     | 1.41  | 103.3 | 102.9 |
| 18  | 0      | 2      | 0     | 0     | 0     | 0     | 100.4 | 100.1 |
| 19  | 0      | 2      | 0     | 0     | 0     | 0     | 100.4 | 100.3 |
| 20  | 0      | 2      | 0     | 0     | 0     | 0     | 100.4 | 100.3 |